Project – mutual inductance circuit

Subject: Numerical methods and simulations
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Methods and Tools

The circuit has three state variables: primary current i_1, secondary current i_2, and capacitor voltage uc. Kirchhoff's voltage laws give:

$$di_1/dt = (1/D_1) \times [-(R_1/M) \cdot i_1 + (R_2/L_2) \cdot i_2 - (1/M) \cdot uc + (1/M) \cdot e]$$

$$di_2/dt = (1/D_2) \times [-(R_1/L_1) \cdot i_1 + (R_2/M) \cdot i_2 - (1/L_1) \cdot uc + (1/L_1) \cdot e]$$

$$duc/dt = (1/C) \cdot i_1$$

$D_1 = L_1/M - M/L_2$, $D_2 = M/L_1 - L_2/M$.

Initial conditions: $i_1 = i_2 = uc = 0$. Parameters: $R_1 = 0.1\,\Omega$, $R_2 = 10\,\Omega$, $C = 0.5\,F$, $L_1 = 3\,H$, $L_2 = 5\,H$, $M = 0.8\,H$. Step h = 0.001 s, t ϵ[0, 30] s.

Part 1 – ODE Solvers

Let $y = [i_1, i_2, uc]^T$ and F(t, y) = right-hand side of the system above.

Euler (O(h)):

The simplest method; it advances the solution by one step using the derivative at the start of the interval.

$$y(i+1) = y(i) + h \times F(t(i), y(i))$$

Heun (O(h²)):

A two-stage improvement over Euler: a preliminary step (k_1) gives a first estimate of the next point, and a corrector step (k_2) evaluates the derivative there; the average of both slopes is used.

$$k_1 = F(t(i), y(i))$$

$$k_2 = F(t(i+1), y(i) + h \cdot k_1)$$

$$y(i+1) = y(i) + h \times (k_1 + k_2)/2$$

RK4 (O(h⁴)):

Uses four slope estimates within each step: at the start, twice at the midpoint, and once at the end, and combines them in a 1:2:2:1 weighted average, giving fourth order accuracy.

$$k_1 = F(t(i), \quad y(i))$$

$$k_2 = F(t(i) + h/2, y(i) + (h/2) \cdot k_1)$$

$$k_3 = F(t(i) + h/2, y(i) + (h/2)\cdot k_2)$$

$$k_4 = F(t(i+1),\ y(i) + h\cdot k_3)$$

$$y(i+1) = y(i) + h \times (k_1 + 2k_2 + 2k_3 + k_4)/6$$

Part 2 – Nonlinear Mutual Inductance

The voltage across L_1 is $u_{L1} = e - uc - R_1 \cdot i_1$. $M_n = M_n(|u_{L1}|)$; clamped to 0.18 H when $|u_{L1}| > 300\,V$. RK4 is used with M replaced by M_n at each step. Four approximation methods:

Lagrange (degree 7):

The curve passes exactly through all eight data points; each basis polynomial $L_i(x)$ equals 1 at its own node x_i and 0 at all other nodes, which guarantees exact interpolation.

$$M_n(x) = \Sigma y_i \cdot L_i(x),\quad L_i(x) = \Pi_{\{j \neq i\}} (x - x_j)/(x_i - x_j)$$

Natural cubic spline:

The data range is divided into segments; a cubic polynomial is fitted to each segment and the pieces are joined so the value, slope, and curvature match at every join point (C^2 continuity). The second derivative is set to zero at both ends (natural boundary condition).

$$S_k(x) = a_k(x - x_k)^3 + b_k(x - x_k)^2 + c_k(x - x_k) + d_k$$

$$S''(x_0) = S''(x_n) = 0$$

Polynomial approx. (n=3 and n=5):

Instead of passing through every point, a single polynomial minimises the sum of squared errors. The best-fit coefficients c are found from the normal equations; data is normalised to [0,1] before fitting to improve numerical stability.

$$(A^T A)c = A^T y$$

Part 3 – Numerical Integration

$P = \int_0^{30} [R_1 \cdot i_1^2 + R_2 \cdot i_2^2]\, dt$. Let $f(t) = R_1 i_1^2 + R_2 i_2^2$. Two methods compared at h = 0.001 s (short) and h = 0.5 s (long):

Composite rectangles (O(h)):

The integrand is approximated by a constant (left-side rectangle) over each sub-interval of width h; all N rectangles are summed.

$$P \approx h \cdot \Sigma_{\{i=0\}}^{\{N-1\}} f(t_i)$$

Composite trapezoidal (O(h²)):

The integrand is approximated by a straight line over each sub-interval; the area of each trapezoid equals h multiplied by the average of its two endpoint values. More accurate than rectangles for the same h.

$$P \approx h \cdot [\, f(t_0)/2 + \Sigma_{\{i=1\}}^{\{N-1\}} f(t_i) + f(t_n)/2\,]$$

Part 4 – Root Finding

$G(f) = P(f) - 406 = 0$, where $e(t) = 100 \cdot sin(2\pi f t)$. Three methods:

Bisection: $c = (a + b)/2$; replace endpoint so root stays in [a,b]; repeat until $|b - a| < \varepsilon$. Safe but slow — the interval halves each step regardless of G's shape.

Secant:

Estimates the slope from the two most recent function values instead of computing the derivative analytically; does not require a starting bracket.

$f(i + 1) = f(i) - G(f(i)) \cdot (f(i) - f(i - 1)) / (G(f(i)) - G(f(i - 1)))$

Quasi-Newton:

Newton's method with a numerically approximated derivative; Δf is chosen so that the estimate changes by less than 1% when Δf is halved. Each iteration needs 2 extra simulator calls.

$f(i + 1) = f(i) - G(f(i))/G'(f(i)); \quad G'(f) \approx [G(f + \Delta f) - G(f)] / \Delta f.$

Plots

Part 1

Excitation 1: e(t) = 1 V (constant)

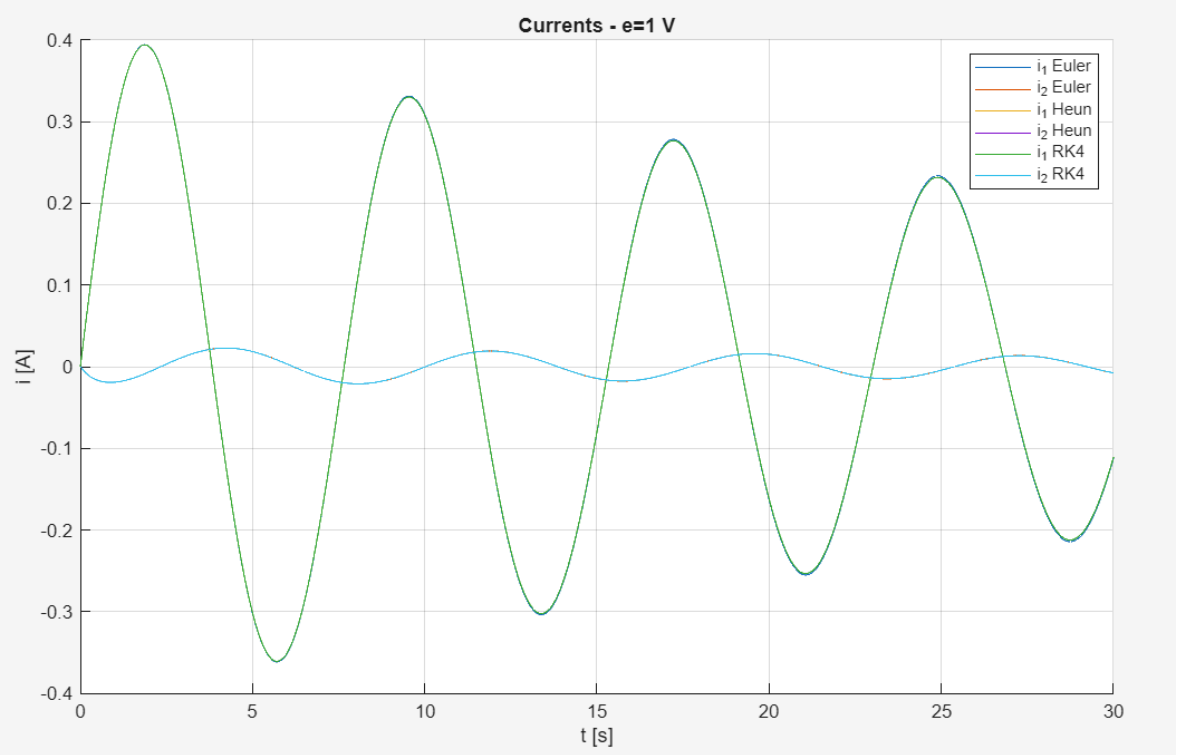


Fig. 1. All three methods produce identical traces at h = 0.001 s; i_1 oscillates with an amplitude of about 0.4 A and slowly decaying envelope, showing the light damping of the circuit.

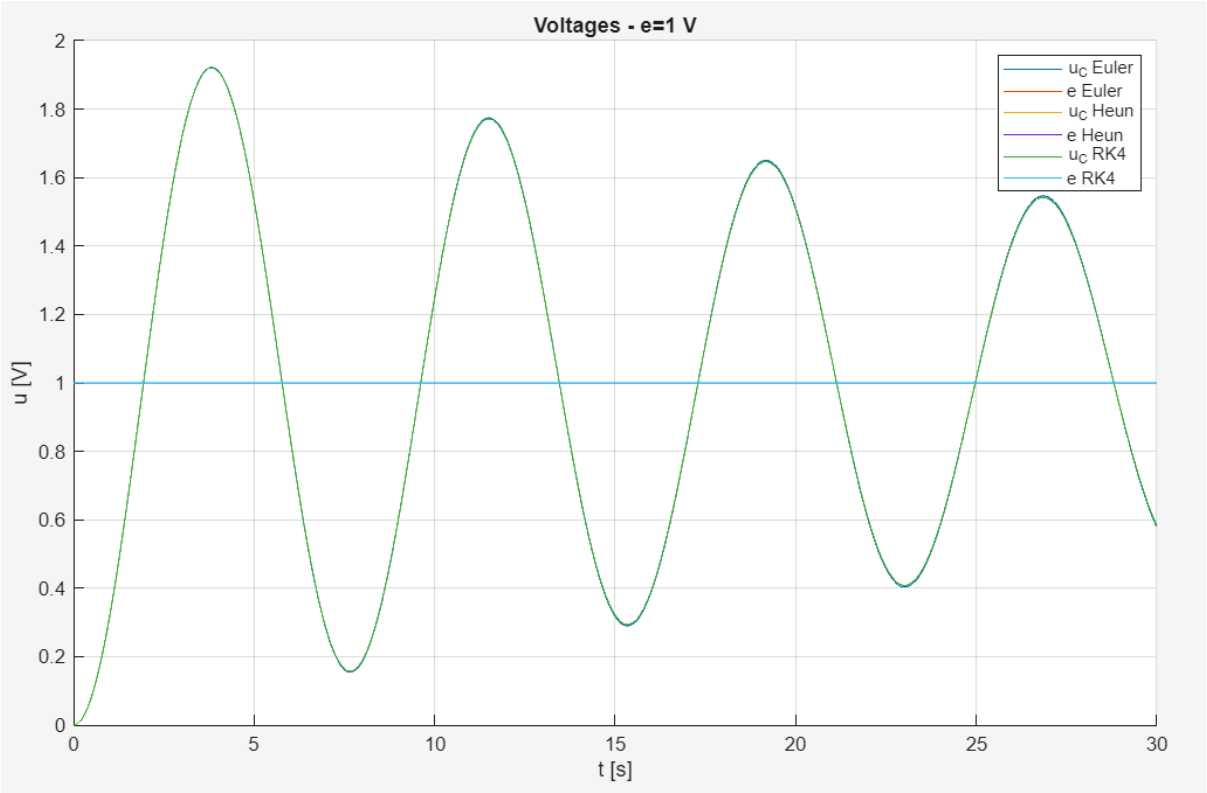


Fig. 2.The capacitor voltage uc oscillates around the constant input e = 1 V with a slowly decreasing amplitude, reflecting the very small energy loss per cycle at $R_1 = 0.1\,\Omega$.

Excitation 2: e(t) = 240 sin(t)

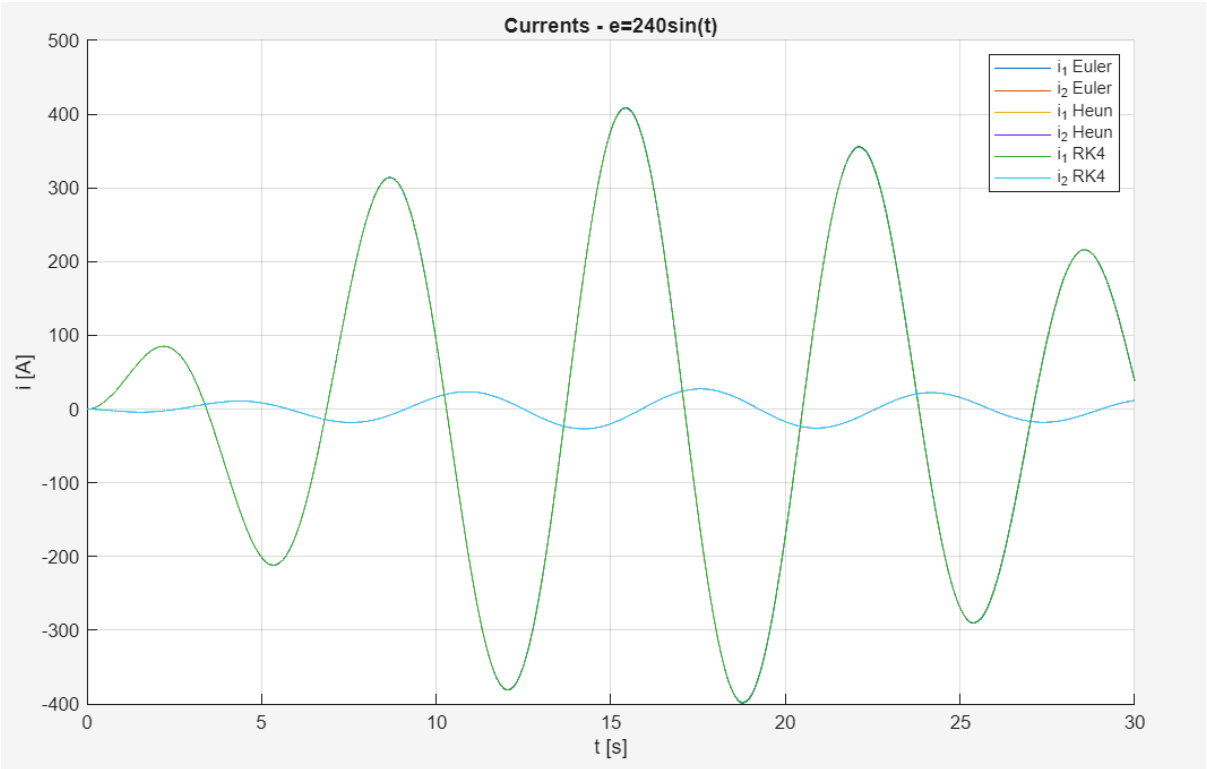


Fig. 3. Near-resonance with $\omega = 1\,rad/s$ causes i_l to reach peak values above 400 A; all three solver traces overlap completely at this step size.

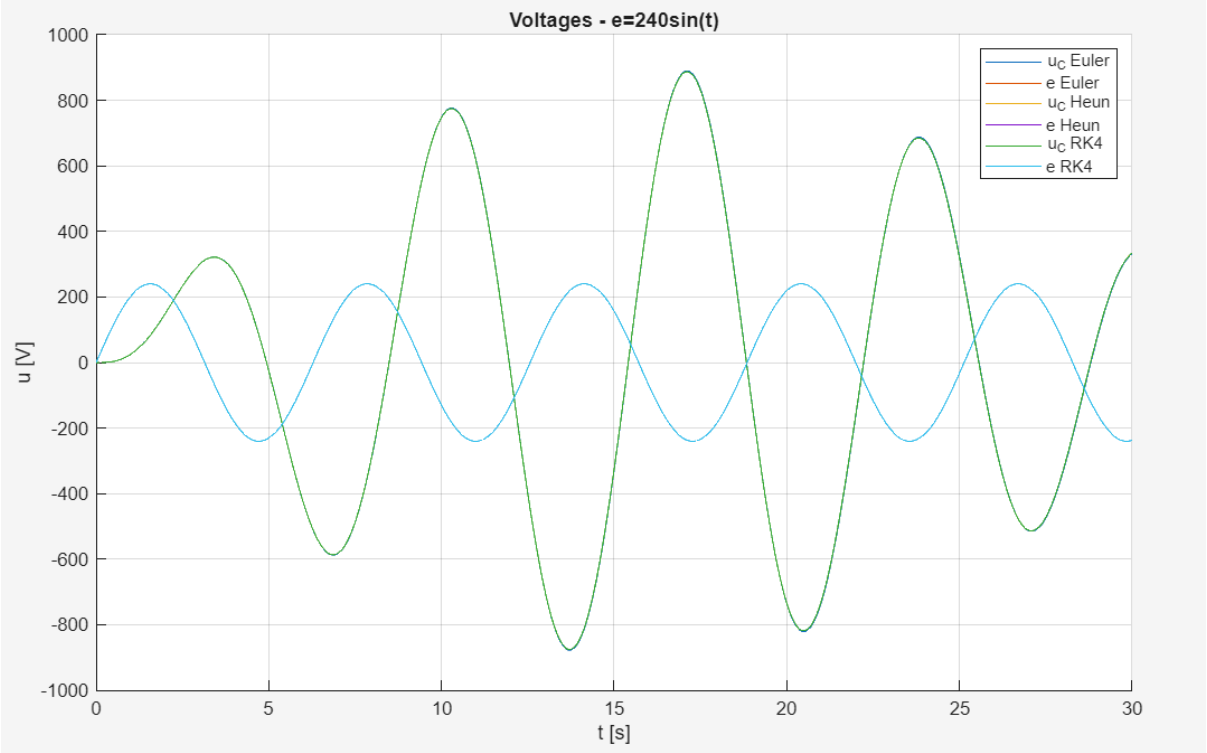


Fig. 4. The capacitor voltage grows steadily in amplitude and approaches $\pm 1000\,V$ as energy accumulates in the near-resonant circuit for e = 240 sin(t).

Excitation 3: e(t) = 210 sin(2πft), f = 5 Hz

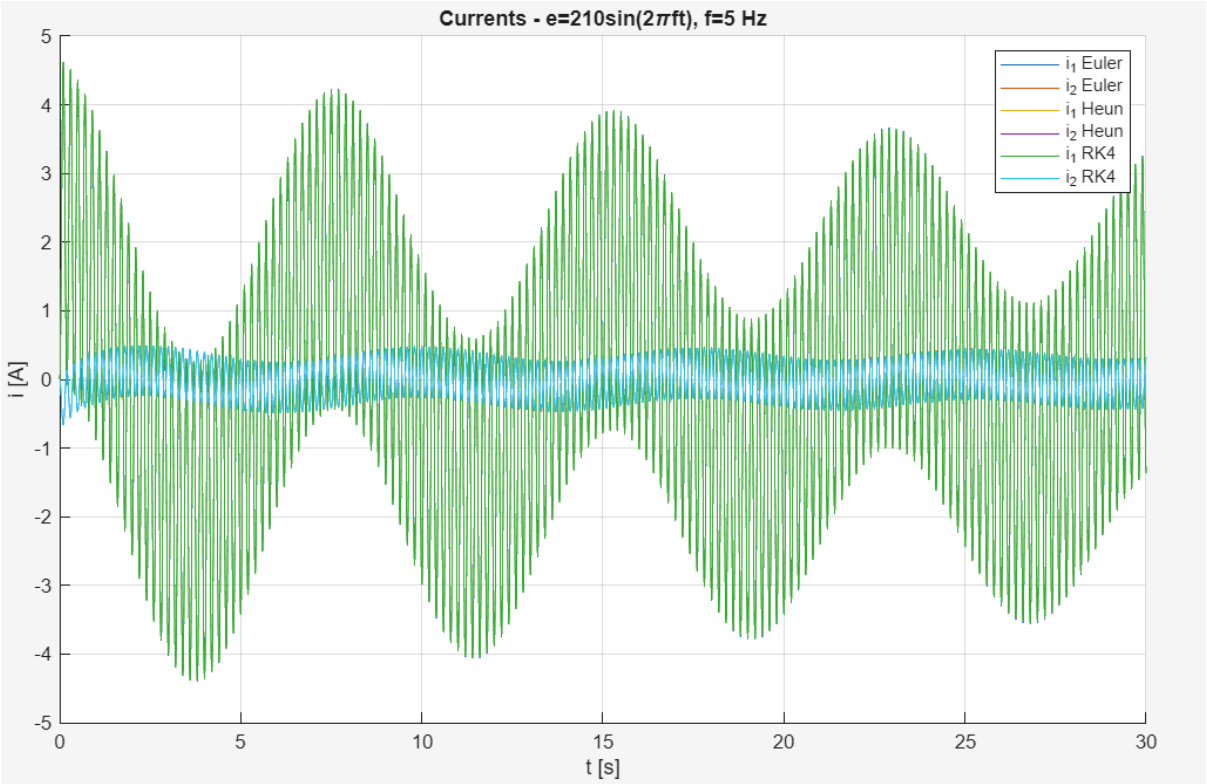


Fig. 5. A slow beat pattern is visible in i_1 due to the interaction between the 5 Hz input and the natural frequency of the circuit; i_2 remains much smaller throughout.

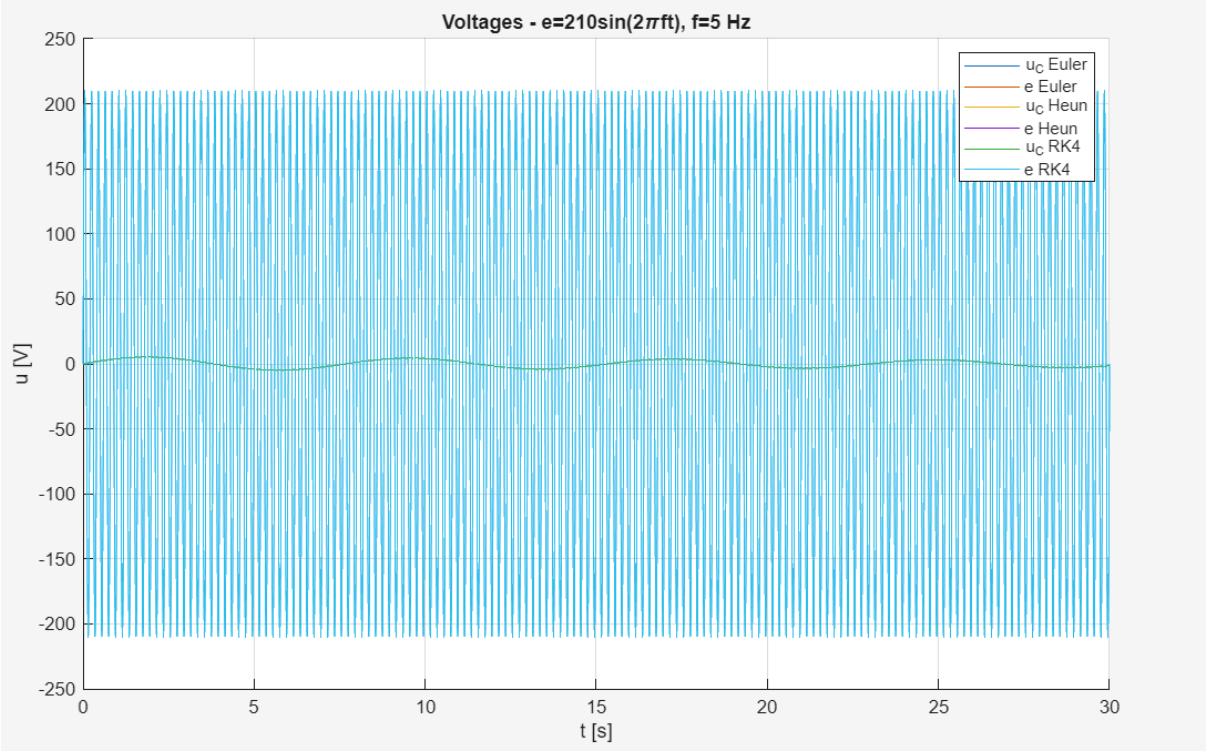


Fig. 6. The capacitor voltage uc changes slowly compared to the 5 Hz input, showing that the circuit acts as a low-pass filter at this frequency.

Excitation 4: e(t) = 120 sin(2πft), f = 50 Hz

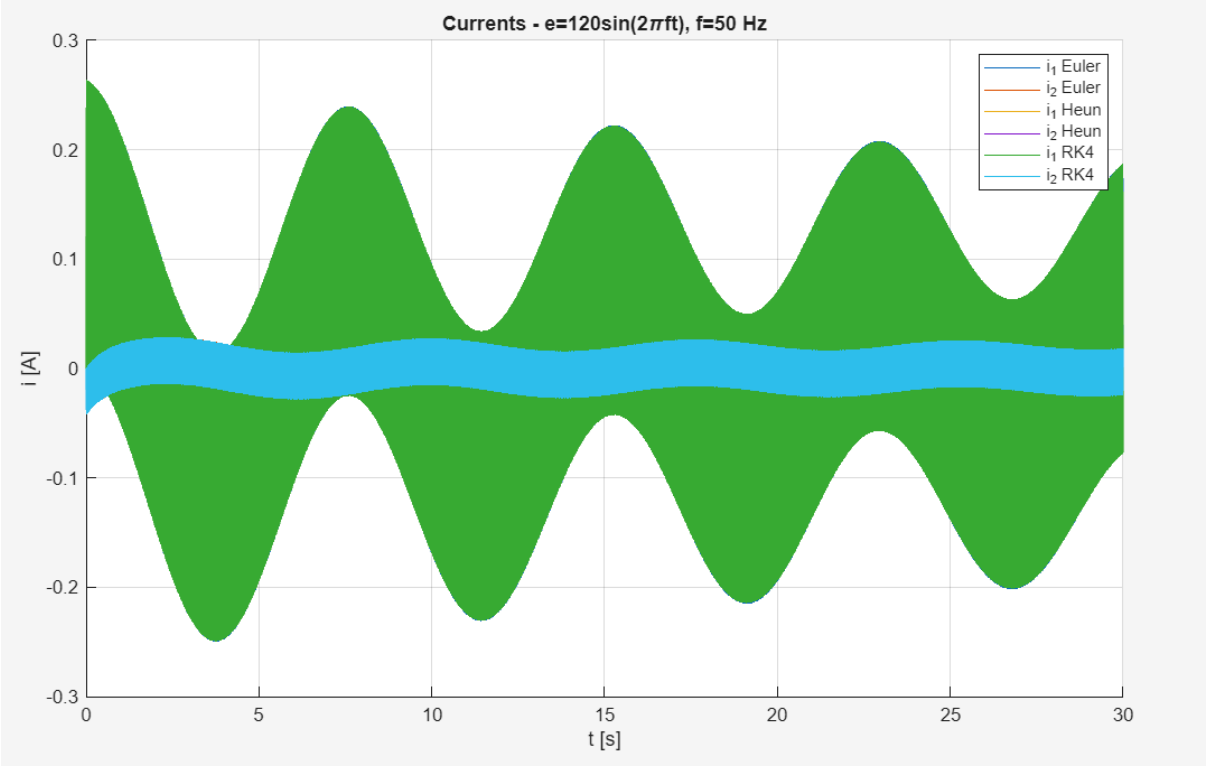


Fig. 7. At 50 Hz both currents show a dense high-frequency oscillation with a slowly modulated envelope; all three methods produce the same traces.

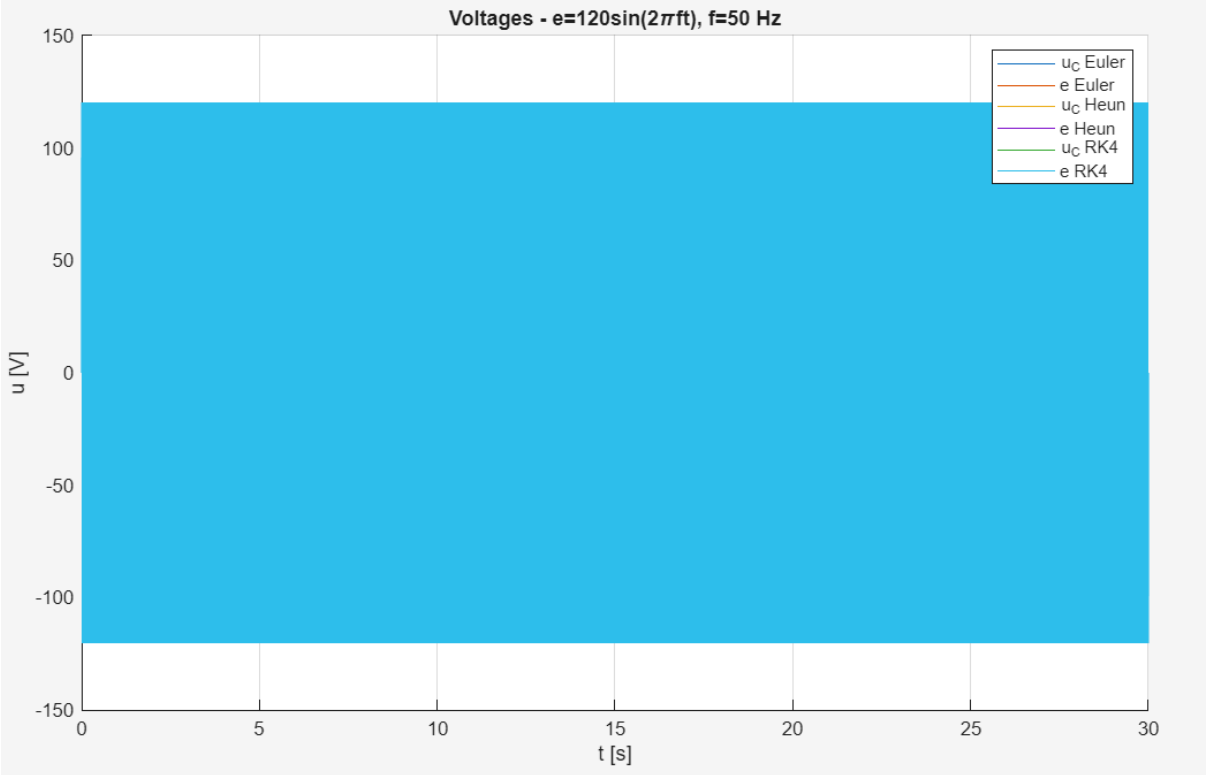


Fig. 8. The input e and uc oscillate at the same frequency and similar amplitude at 50 Hz, where the capacitive impedance is small; all six traces overlap.

Excitation 5: e(t) = 120 square wave, T = 3 s

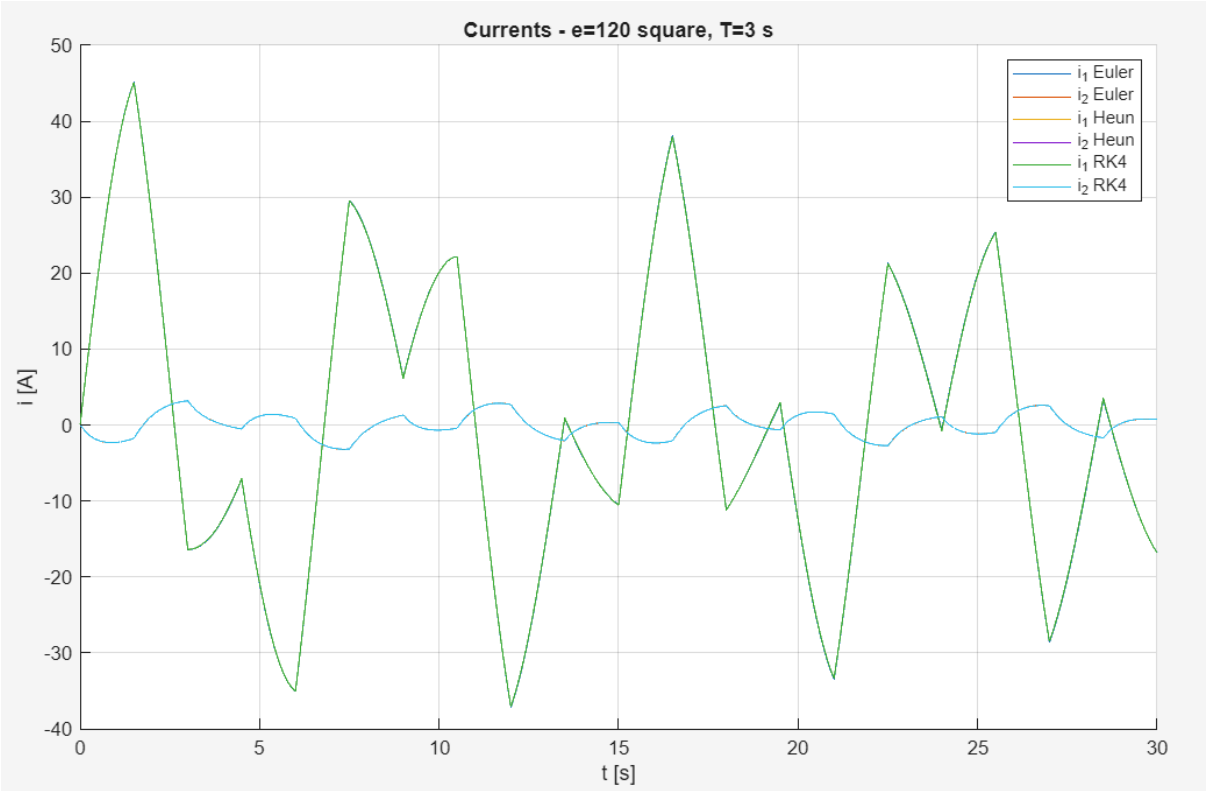


Fig. 9. Each rising and falling edge of the square wave triggers a fast decaying transient in i_1; the three methods agree completely at h = 0.001 s.

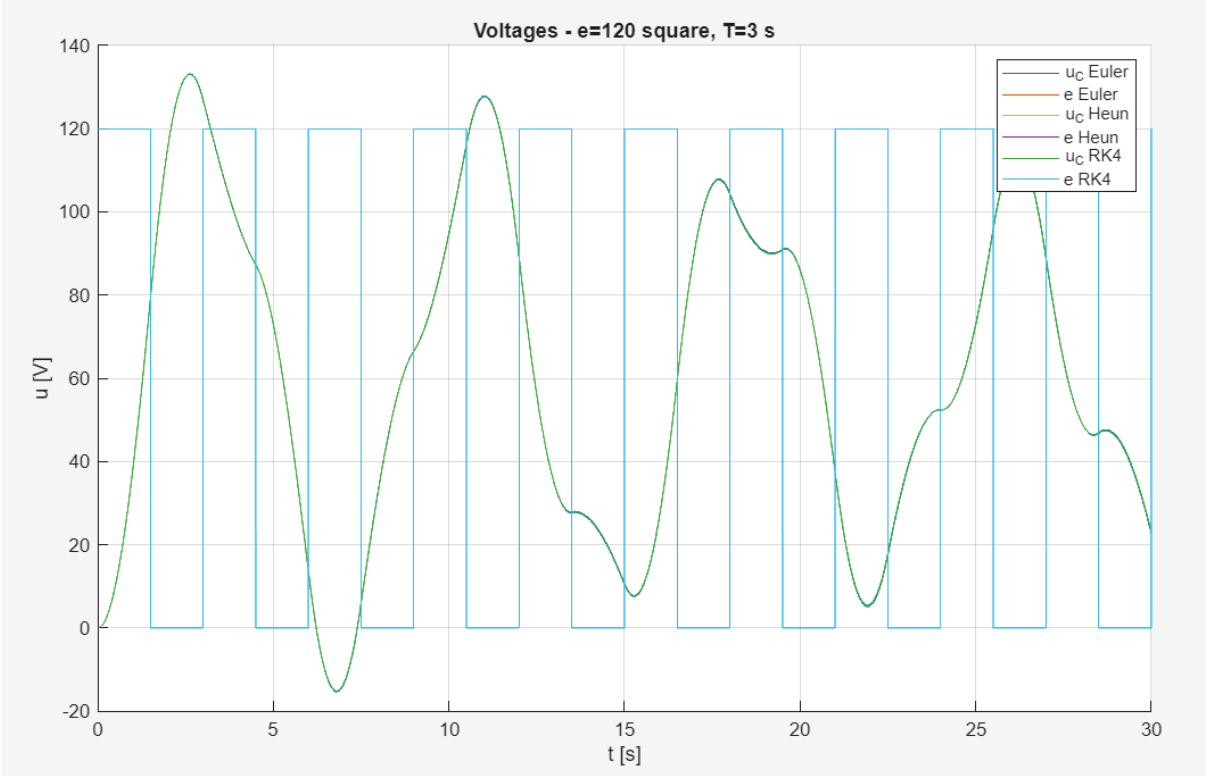


Fig. 10. The capacitor voltage rises toward 120 V during the ON half-period and returns during the OFF half-period, following the square-wave pattern with a lag.

Interpretation

At h = 0.001 s all three methods produce identical traces for every excitation. For e = 1 V, i_1 oscillates slowly with decreasing amplitude due to the small damping ($R_1 = 0.1\ \Omega$). For e = 240 sin(t), near-resonance causes the amplitudes of i_1 and uc to grow large over time. At f = 5 Hz a beat pattern appears in i_1; at f = 50 Hz the high-frequency impedance reduces the currents and uc stays small. For the square-wave input, each switching edge produces a fast decaying transient. If a larger h were used, accuracy differences between the methods would become visible in order Euler < Heun < RK4.

Part 2

Excitation 1: e = 1 V

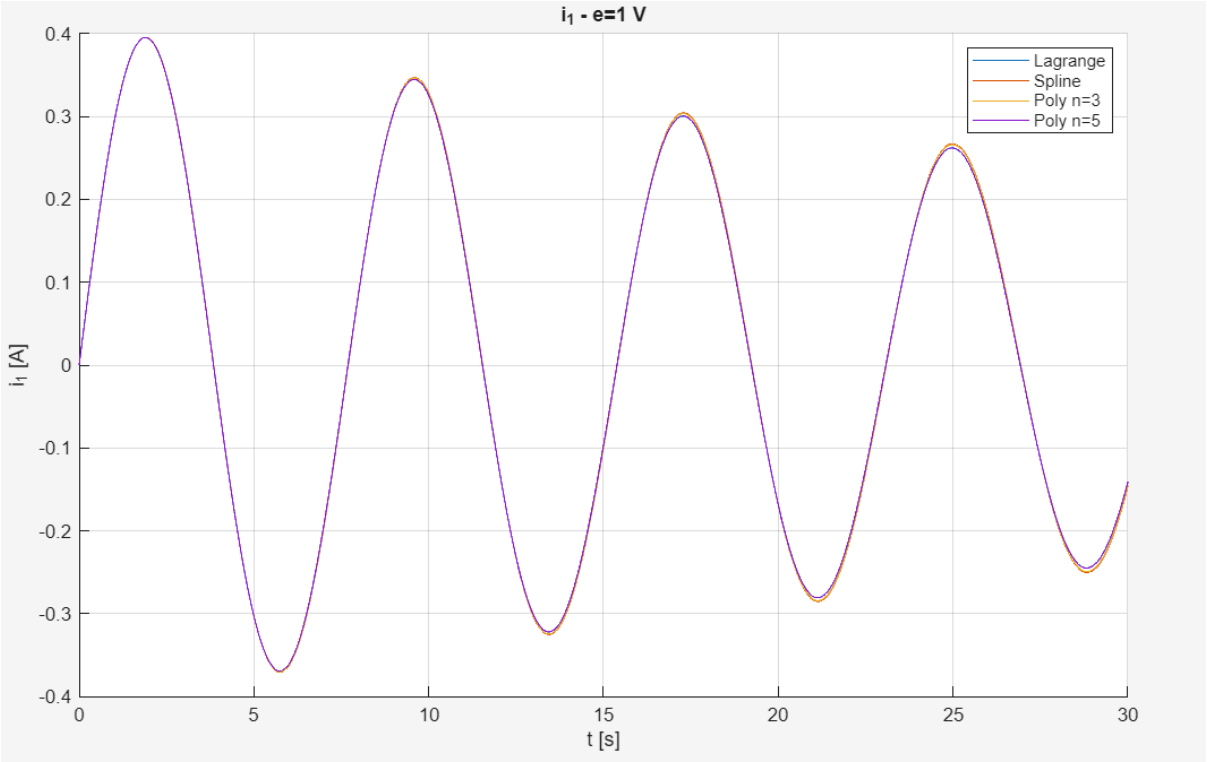


Fig. 11. For the constant source e = 1 V all four M_n methods give the same i_1 trace because u_{L1} stays in the low-voltage part of the data range.

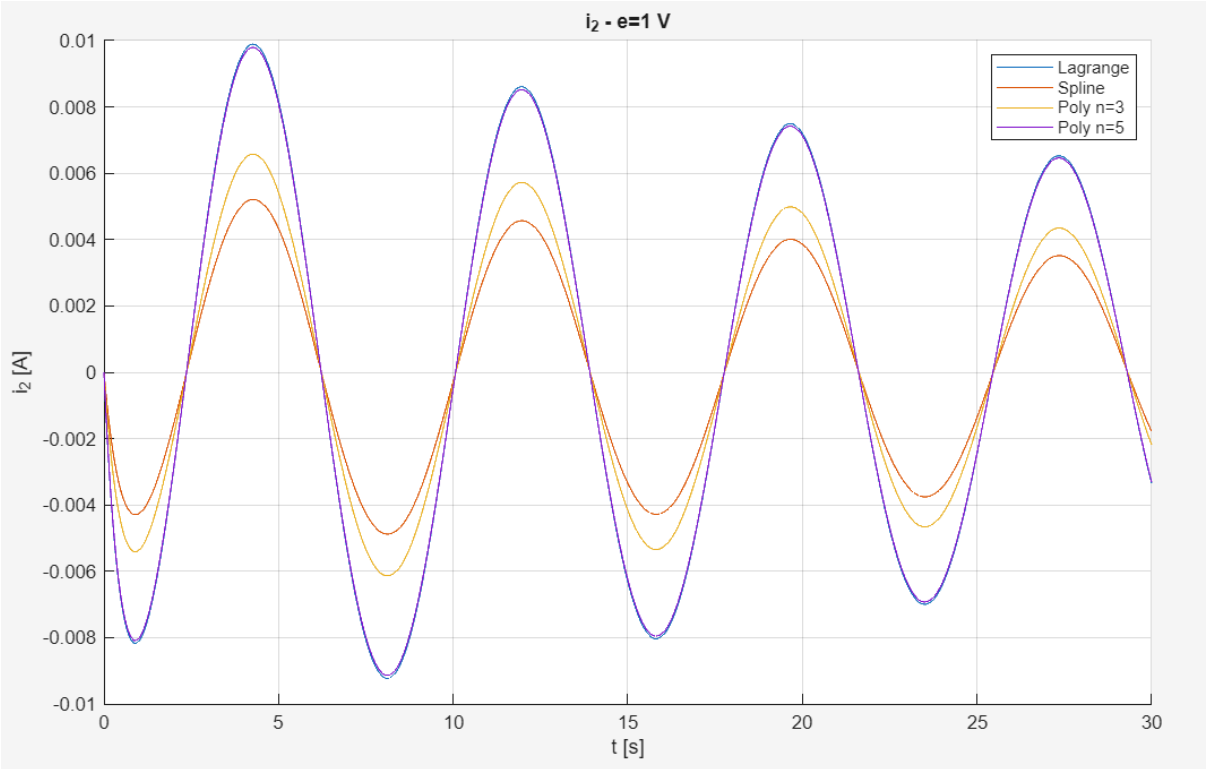


Fig. 12. Small differences in M_n lead to visible differences in i_2 amplitude; Lagrange and spline give higher values than the polynomial approximations.

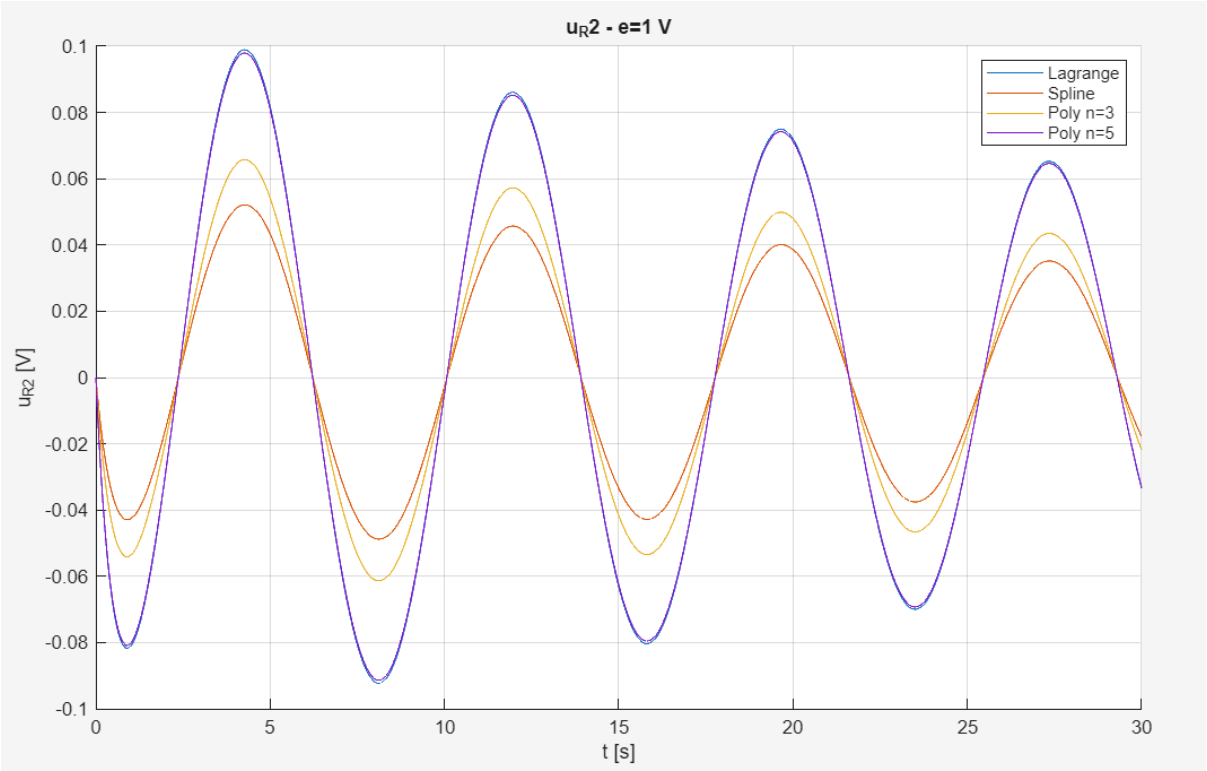


Fig. 13. The voltage $uR_2 = R_2 \cdot i_2$ follows i_2 directly, so the method-to-method differences in i_2 are also visible here.

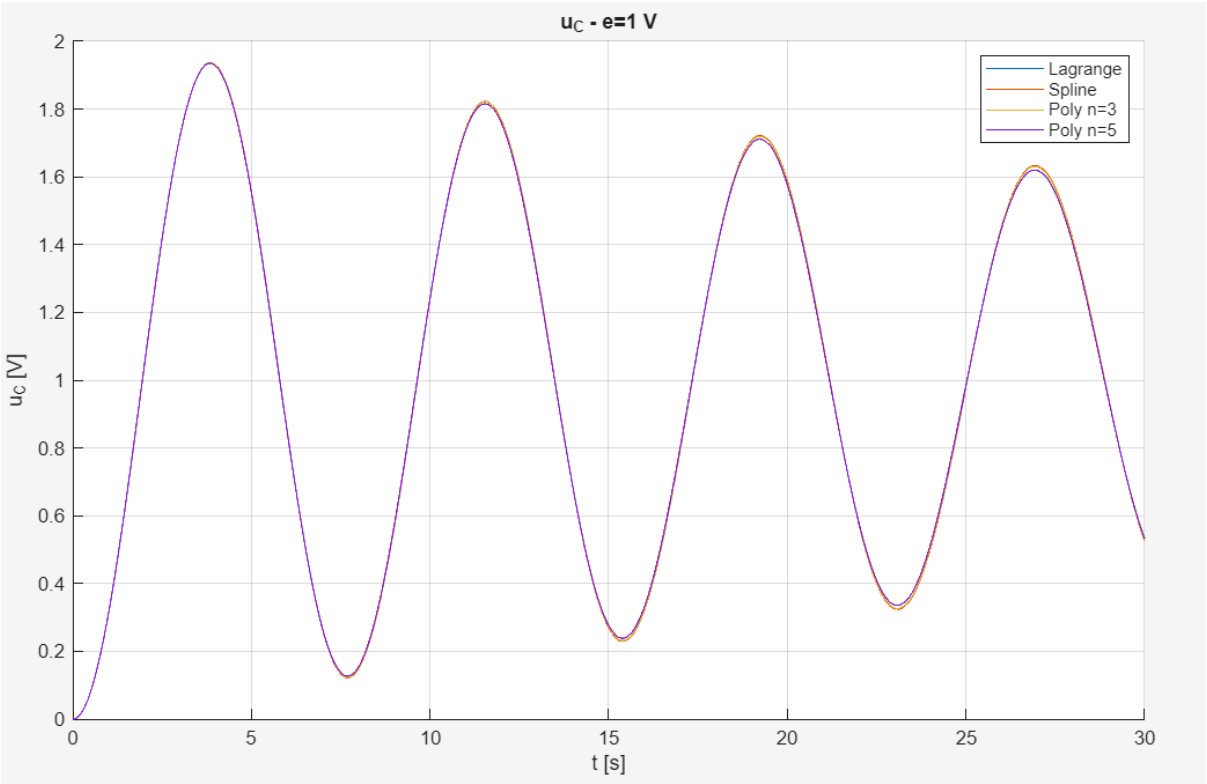


Fig. 14. The capacitor voltage uc is nearly identical for all four methods, confirming that M_n choice mainly affects the secondary circuit.

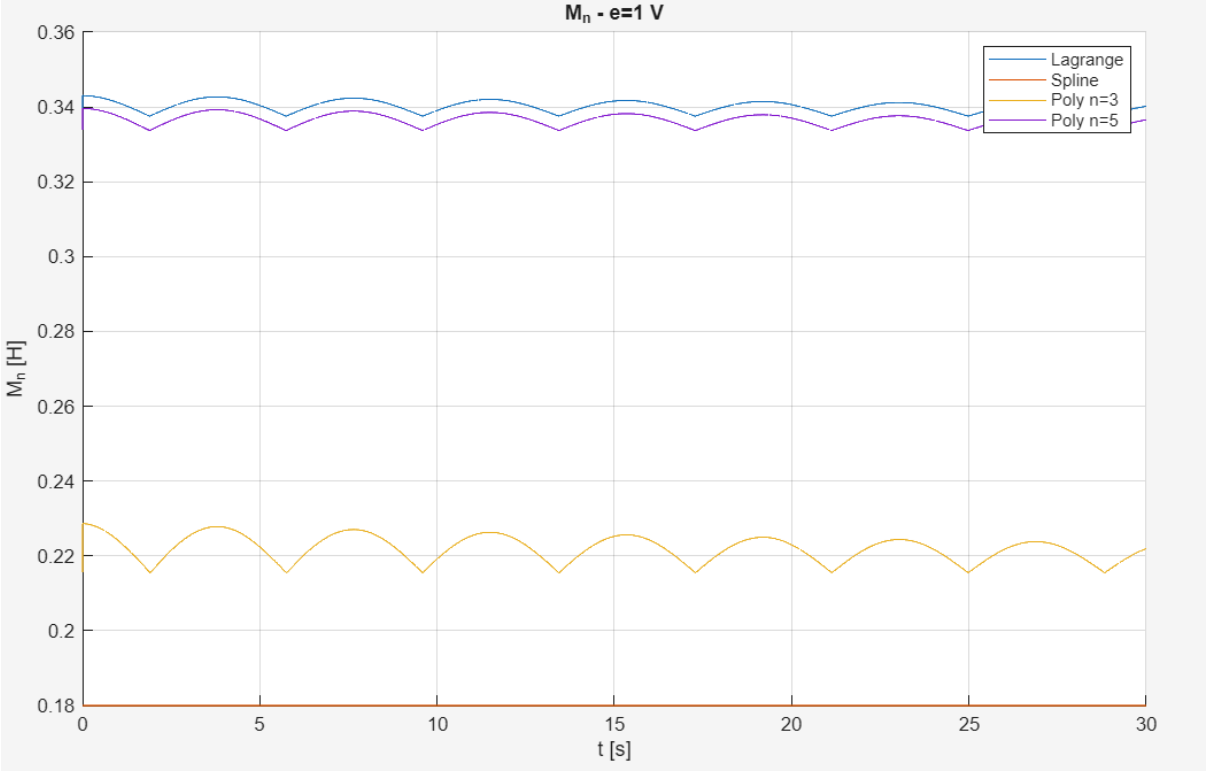


Fig. 15. M_n stays in the range 0.18–0.36 H for e = 1 V; the polynomial approximations assign slightly lower values than Lagrange and spline, which explains the lower i_2 amplitudes.

Excitation 2: e = 240 sin(t) V

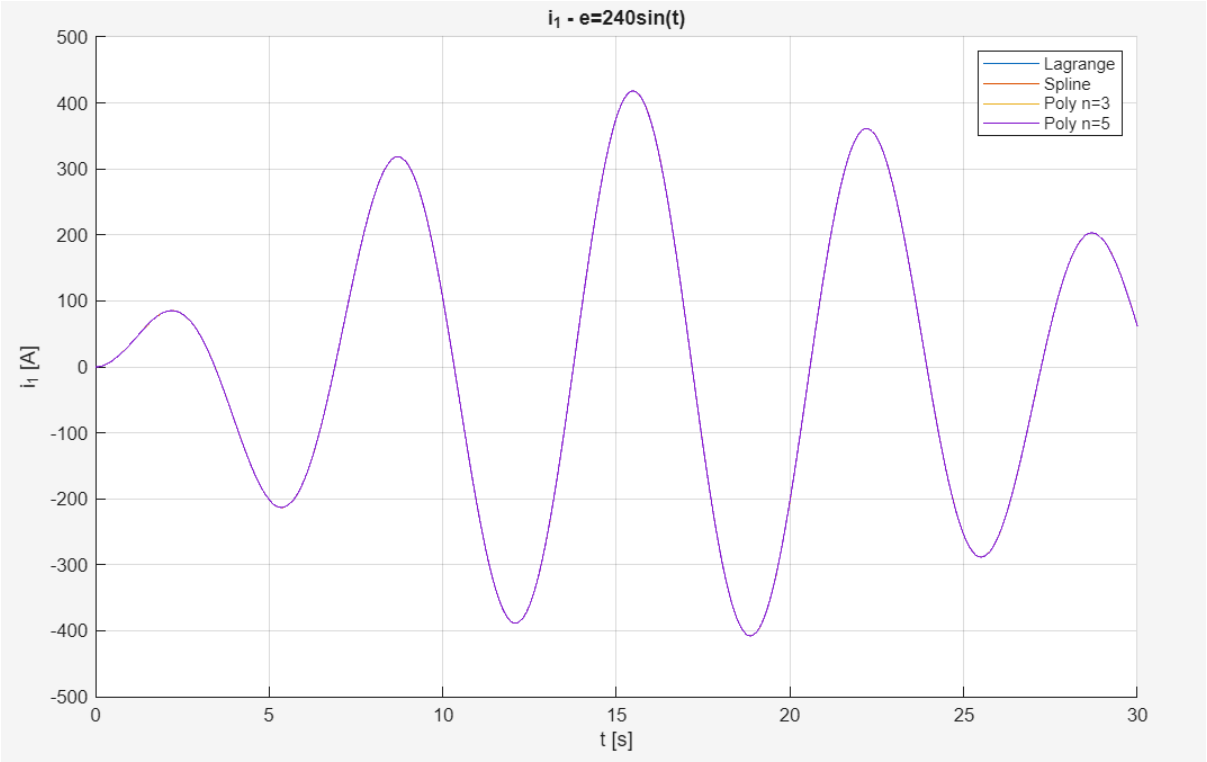


Fig. 16. For e = 240 sin(t), all four methods give almost the same i_1 because the large primary current is dominated by the resonance, not the exact M_n value.

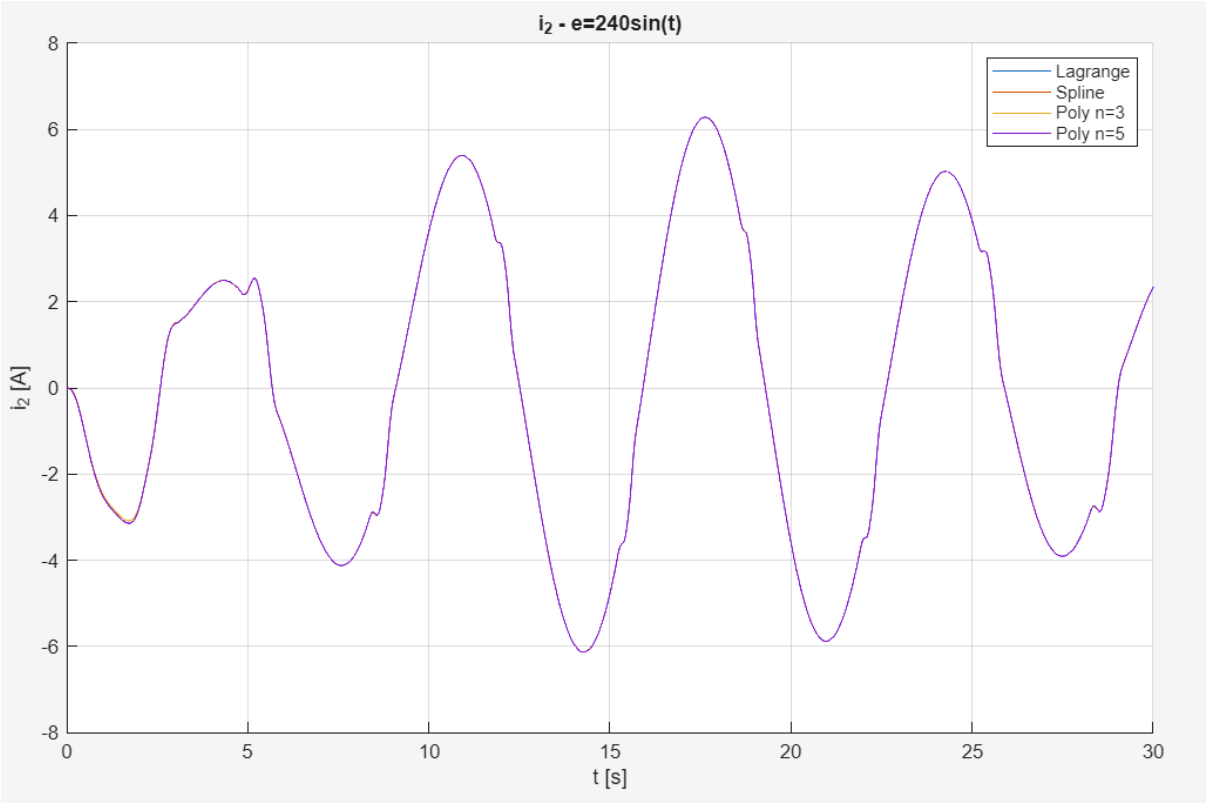


Fig. 17. Small differences between M_n methods are visible in i_2 when u_{L1} is large, but all curves follow the same overall growing envelope.

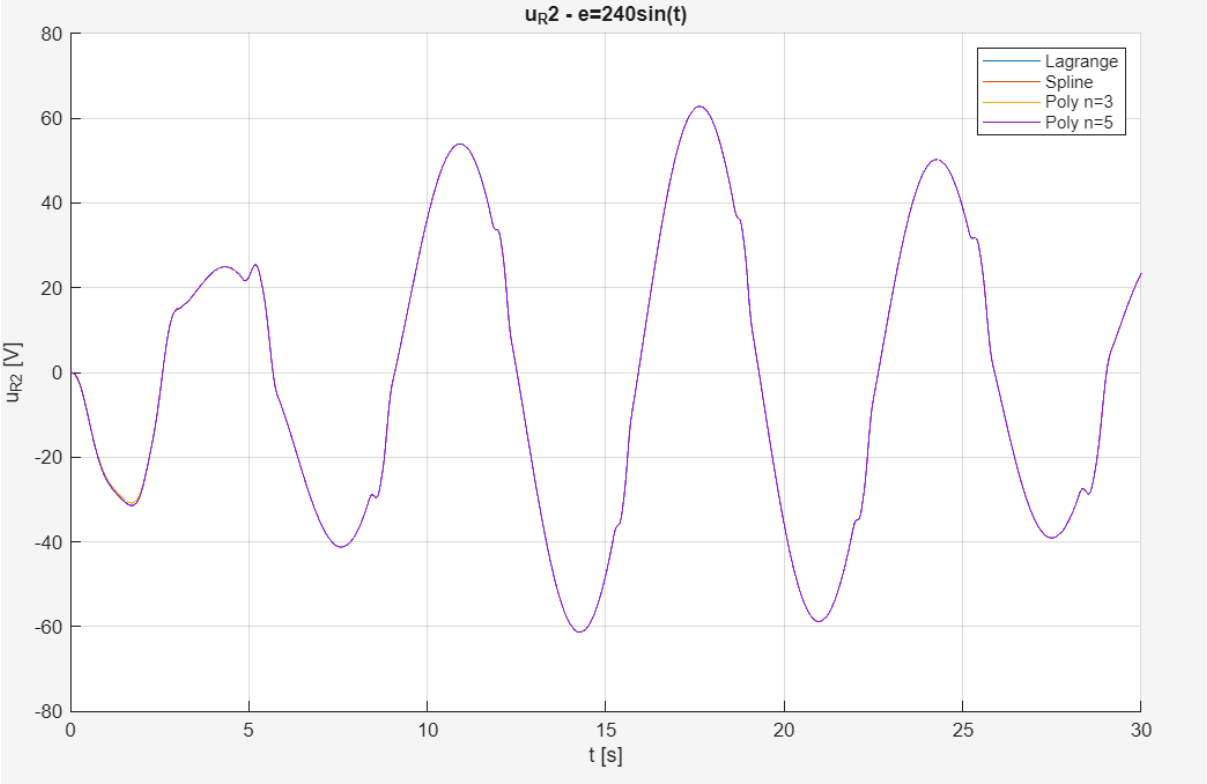


Fig. 18. The voltage uR_2 grows in amplitude over time in line with i_2 at near-resonance conditions; the four methods show the same small spread as in i_2.

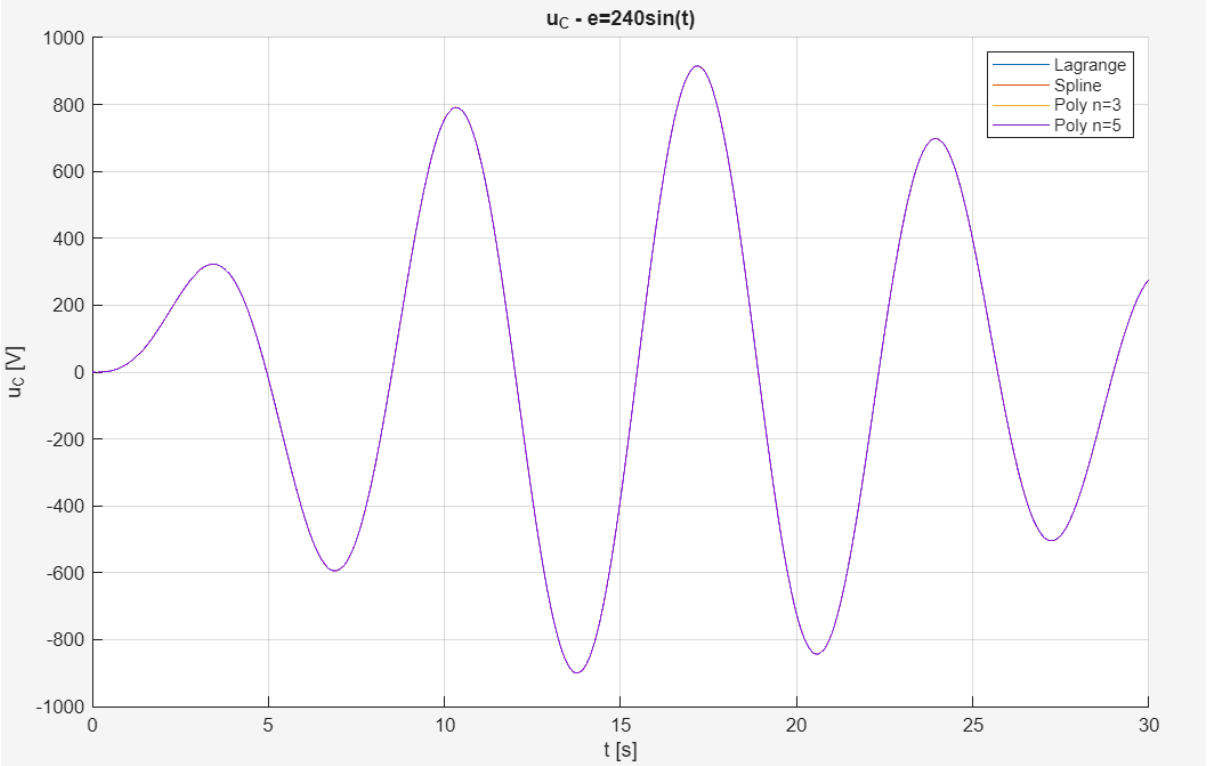


Fig. 19. The capacitor voltage approaches $\pm 1000\,V$ under near-resonance; all four M_n approximations agree closely, showing that uc is not sensitive to the M_n choice.

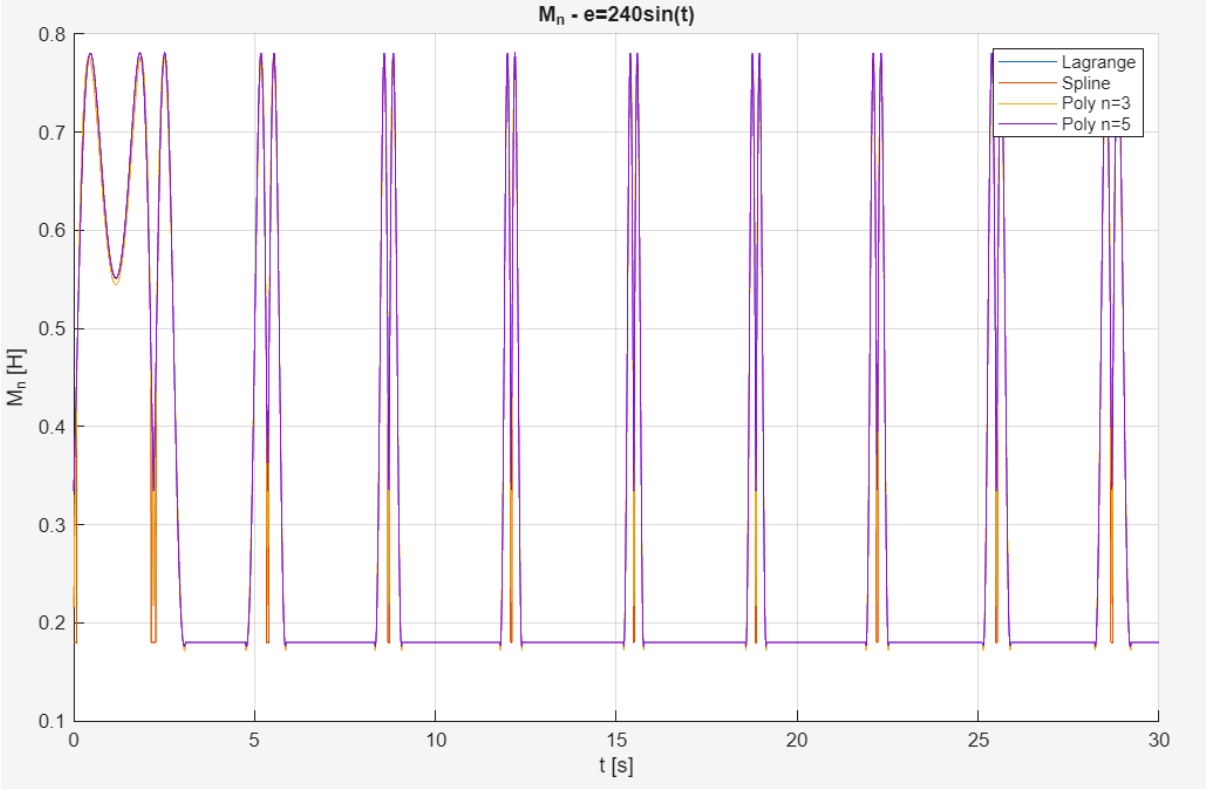


Fig. 20. M_n switches sharply between its minimum and maximum values as u_{L1} sweeps across the data range; small differences between methods are visible at the peaks.

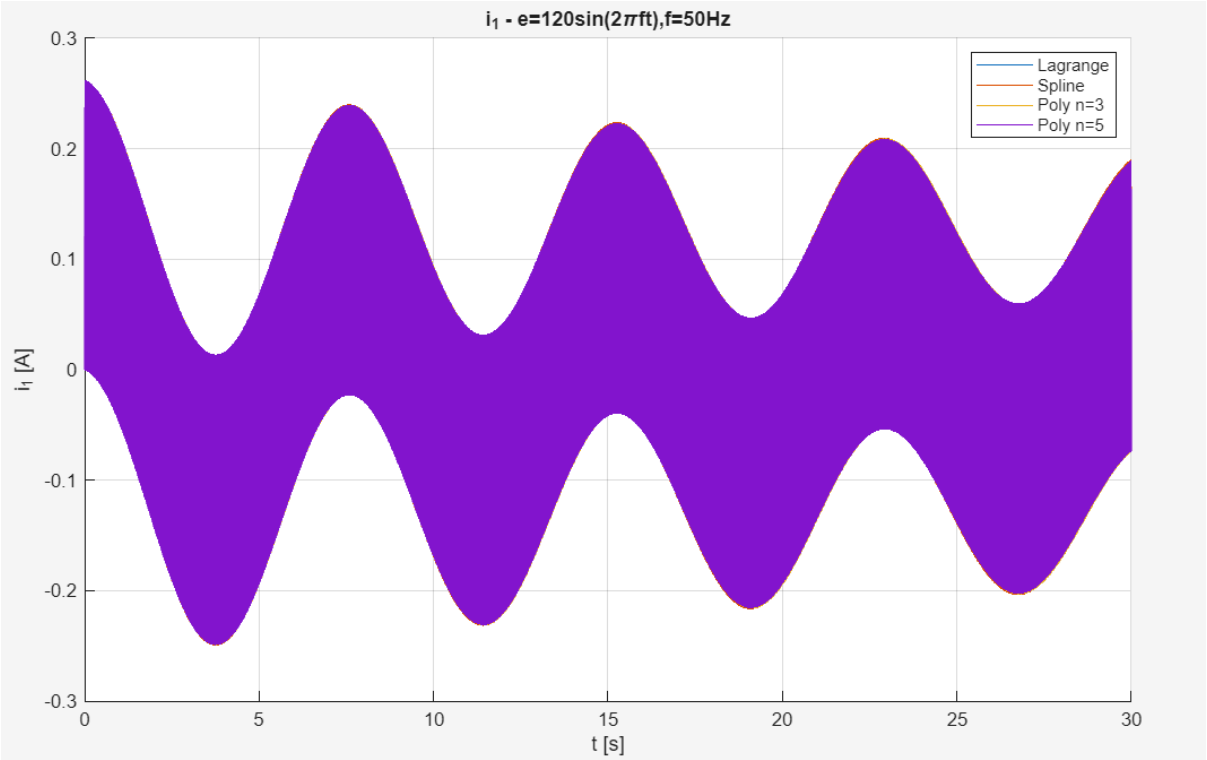


Fig. 21. For $e = 120\,sin(2\pi ft)$ at f = 50 Hz all four methods give the same i_1; the overlapping dense band confirms agreement over the full 30 second simulation.

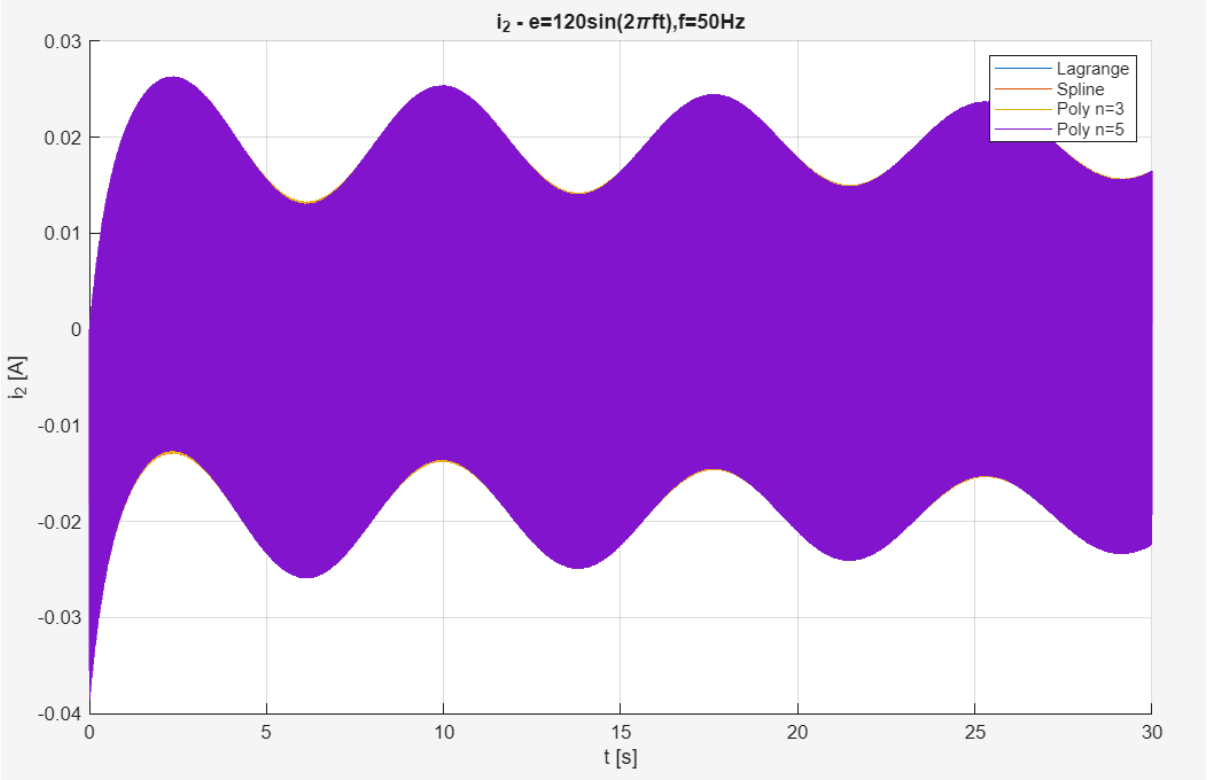


Fig. 22. The secondary current i_2 at 50 Hz is small and grows slowly; all four M_n methods produce closely matching curves.

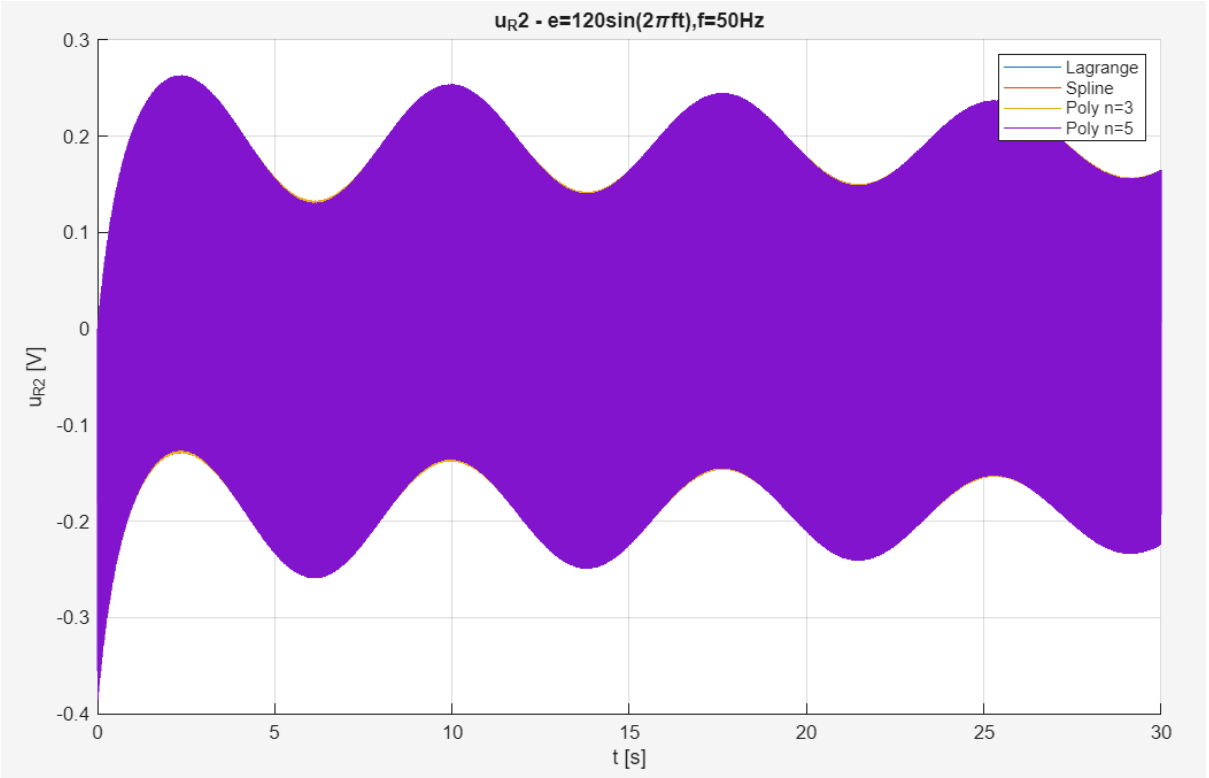


Fig. 23. The voltage uR_2 at 50 Hz follows i_2 directly; the slow amplitude growth is consistent across all four methods.

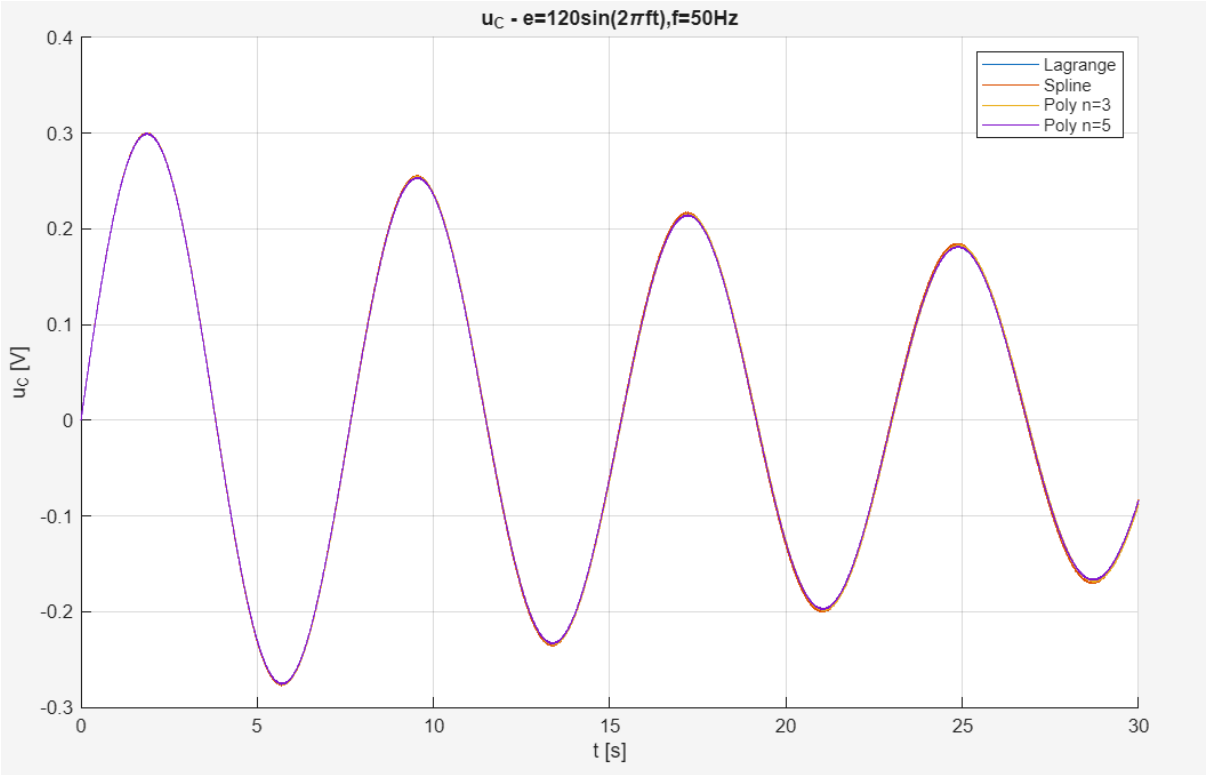


Fig. 24. The capacitor voltage uc shows slow low-amplitude oscillations at 50 Hz; all four M_n methods produce the same trace.

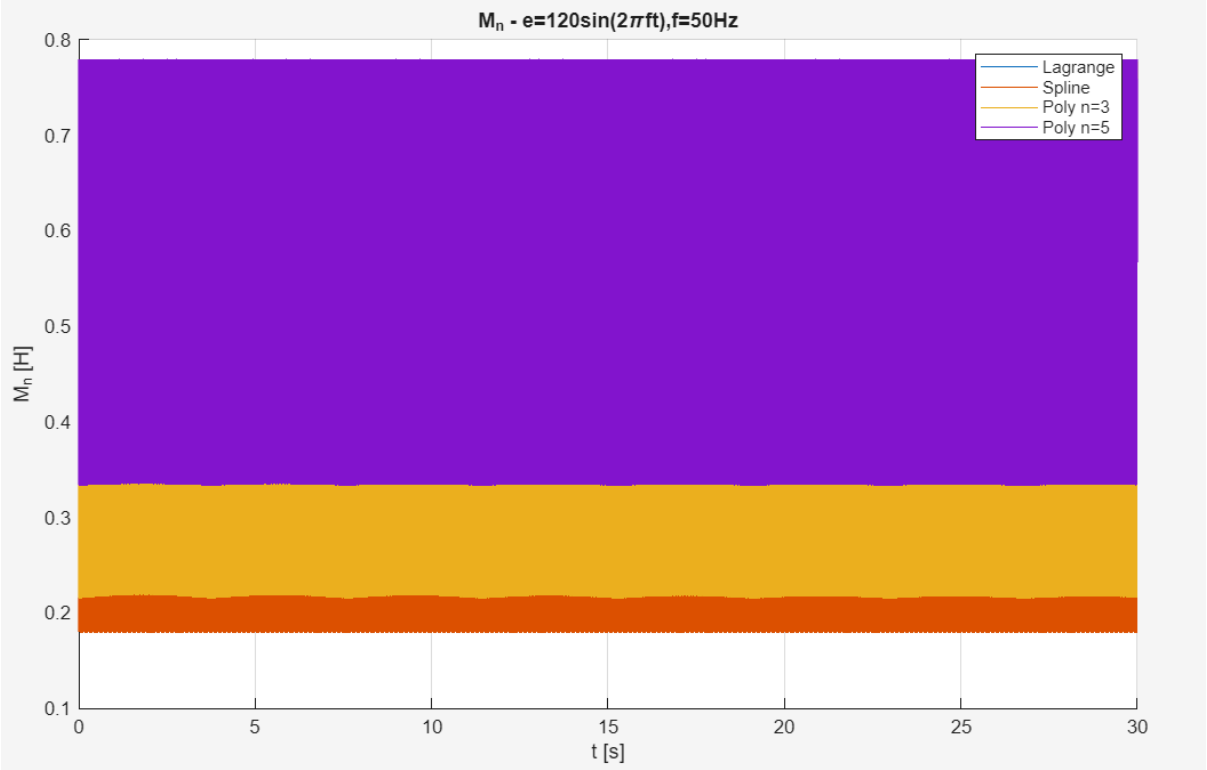


Fig. 25. At 50 Hz, M_n varies widely between methods because large u_{L1} values probe different parts of the data range, yet these differences have little effect on the circuit quantities.

Interpretation

For e = 1 V, u_{L1} stays at low voltages and all four methods give the same M_n values and circuit traces. For e = 240 sin(t), u_{L1} reaches larger values; the Lagrange polynomial and the degree 5 approximation show small unwanted oscillations near the edges of the data range, while the cubic spline and the degree 3 polynomial remain smooth. These differences in M_n cause small variations in i_2 but have little effect on i_1 and uc. For $e = 120\,sin(2\pi ft)$ at f = 50 Hz all four methods agree closely throughout the simulation.

Part 3

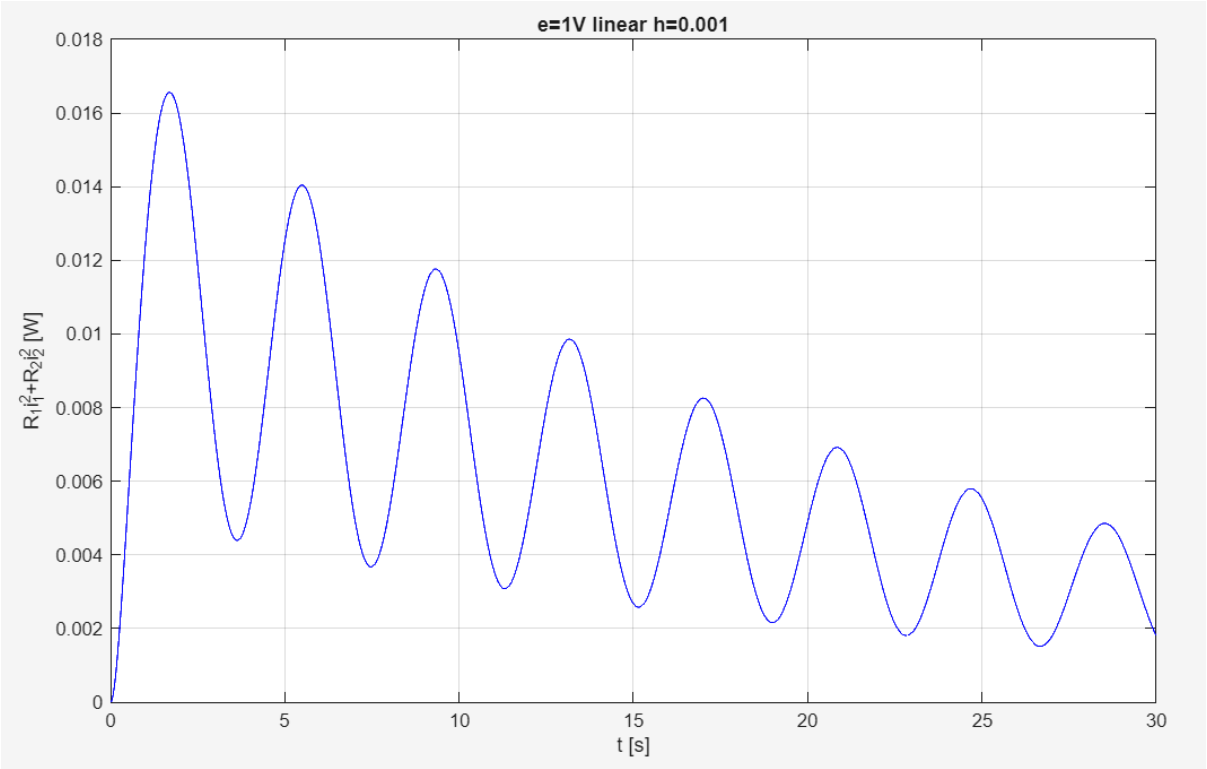


Fig. 26. The instantaneous power $f(t) = R_1 i_1^2 + R_2 i_2^2$ decreases slowly with a small oscillation; the smooth shape makes it easy to integrate accurately with either method.

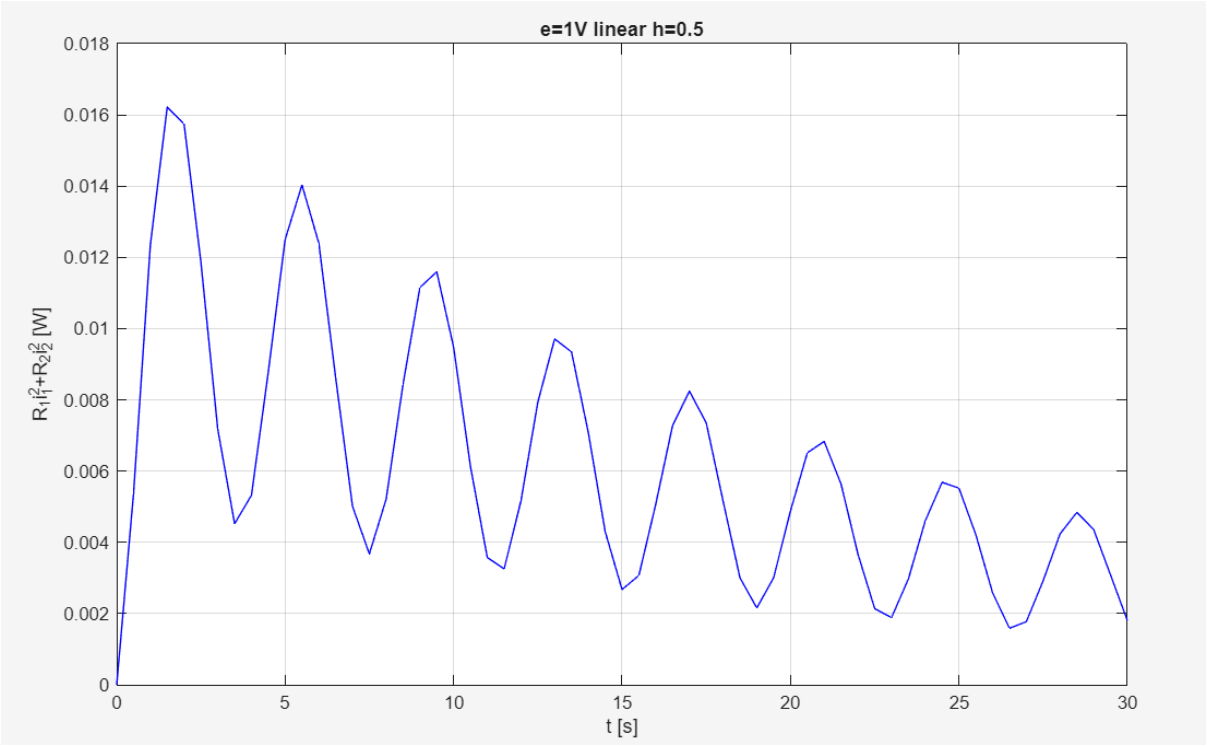


Fig. 27. With the long step h = 0.5 s and linear M, the power curve has fewer samples but the shape is preserved and the total energy agrees closely with the short-step result.

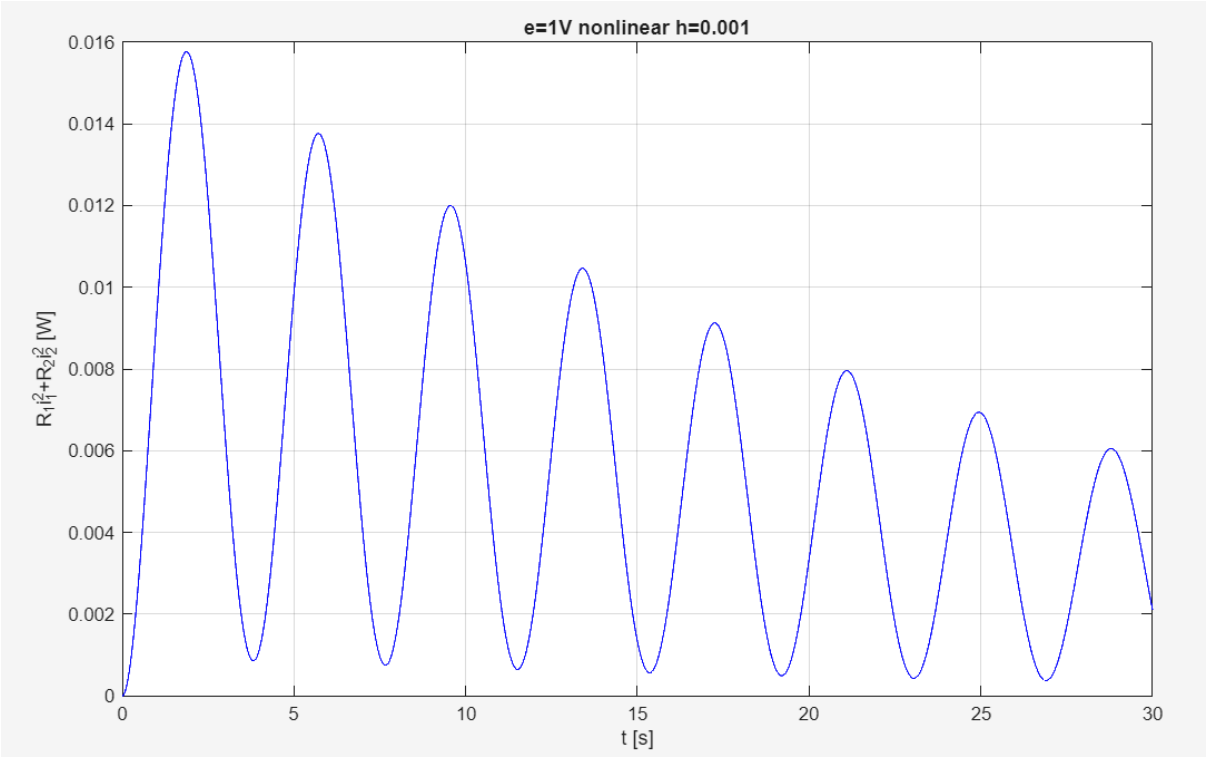


Fig. 28. With nonlinear M and short step, the power amplitude is slightly lower than the linear case, consistent with the lower total energy dissipated under the nonlinear model.

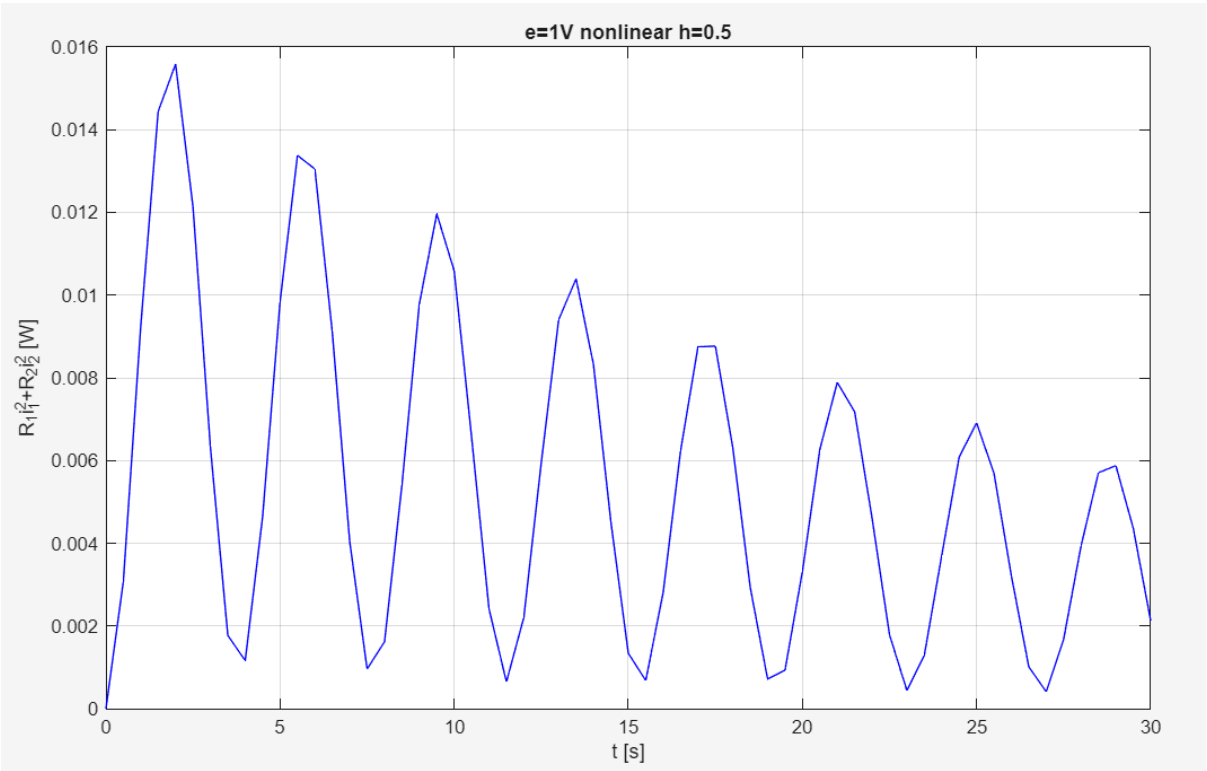


Fig. 29. With nonlinear M and long step h = 0.5 s, the power curve is coarser but the total energy matches the short-step nonlinear result within a small error.

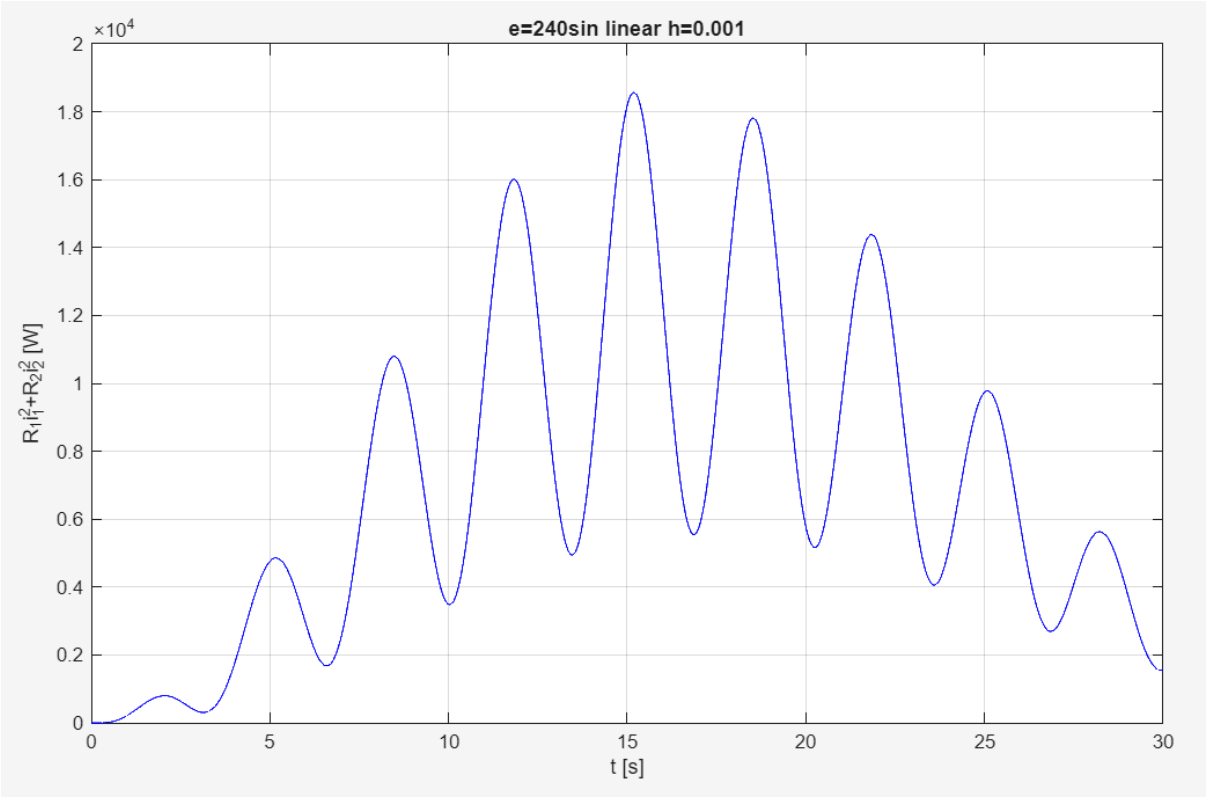


Fig. 30. For e = 240 sin(t) and linear M, the power grows strongly over time due to near-resonance; the smooth envelope means both integration methods give similar totals.

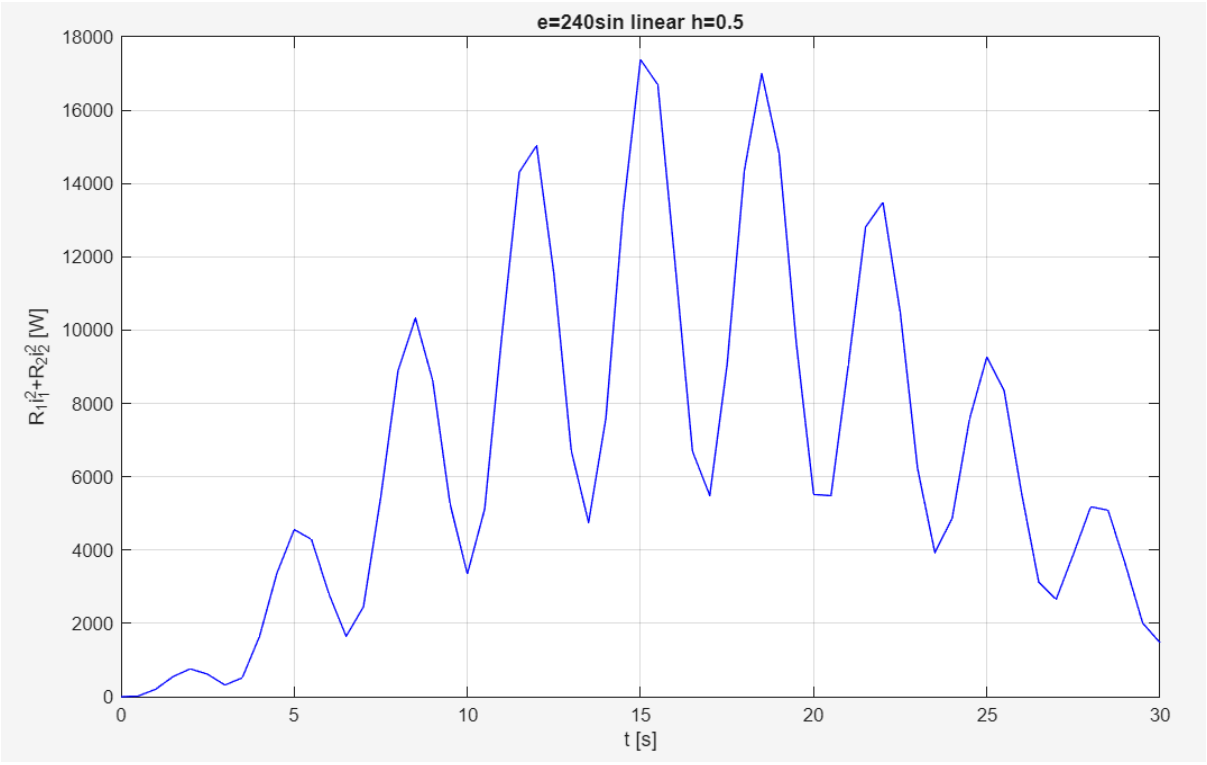


Fig. 31. With the long step and e = 240 sin(t), the peak shapes differ from the short-step case, which accounts for the approximately 5% reduction in total energy.

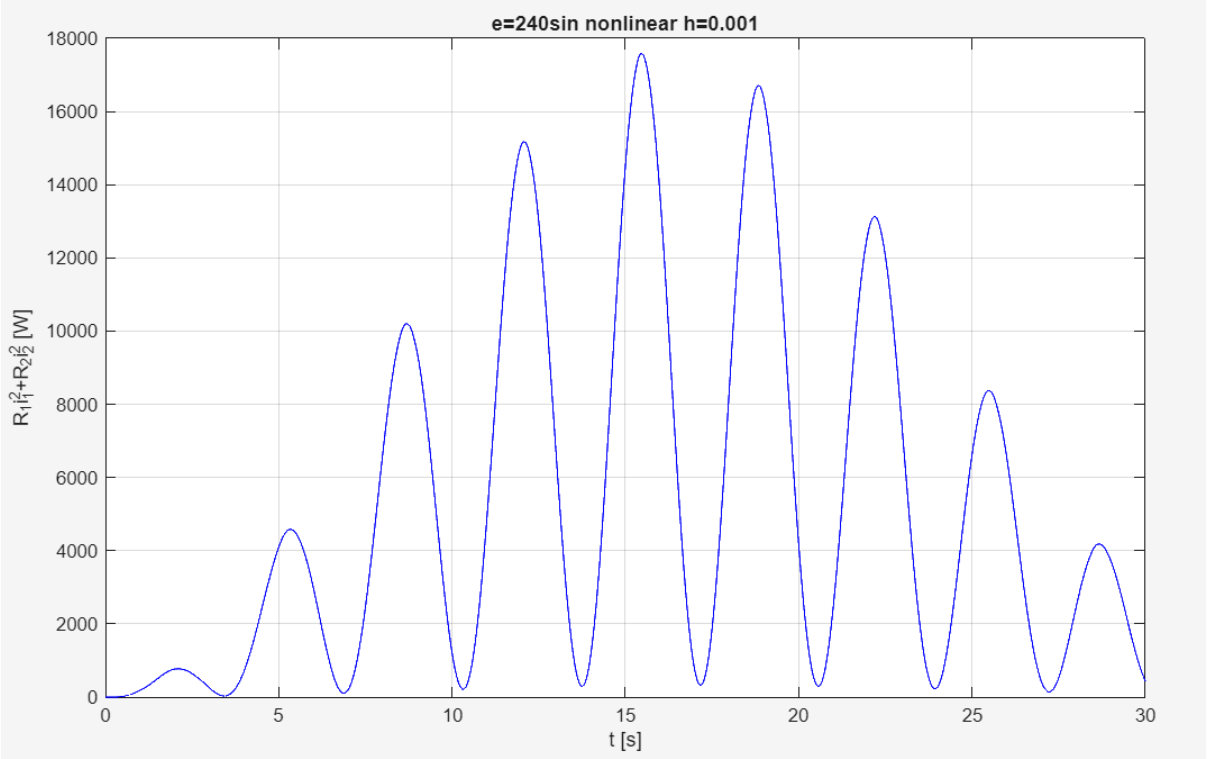


Fig. 32. The power for e = 240 sin(t) with nonlinear M follows the same growing resonance pattern but at lower amplitude, consistent with the reduced energy compared to the linear model.

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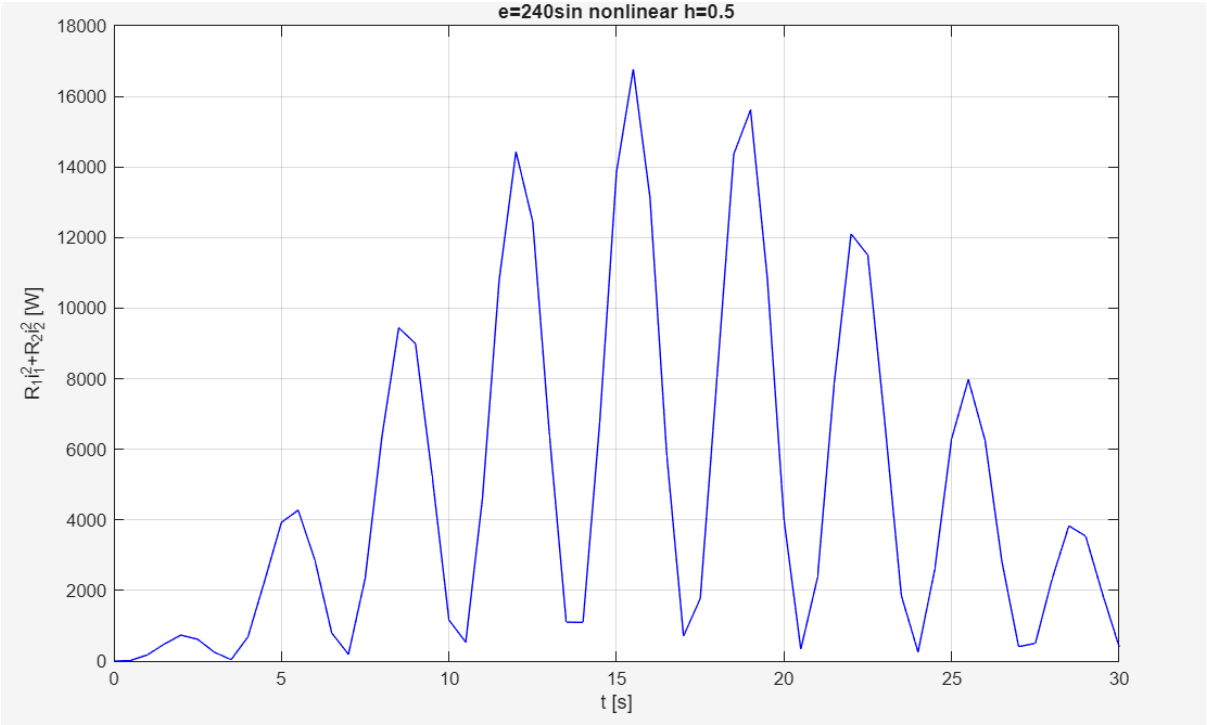


Fig. 33. With nonlinear M and long step, the growing resonance envelope is still visible; the total energy is close to the short-step nonlinear value.

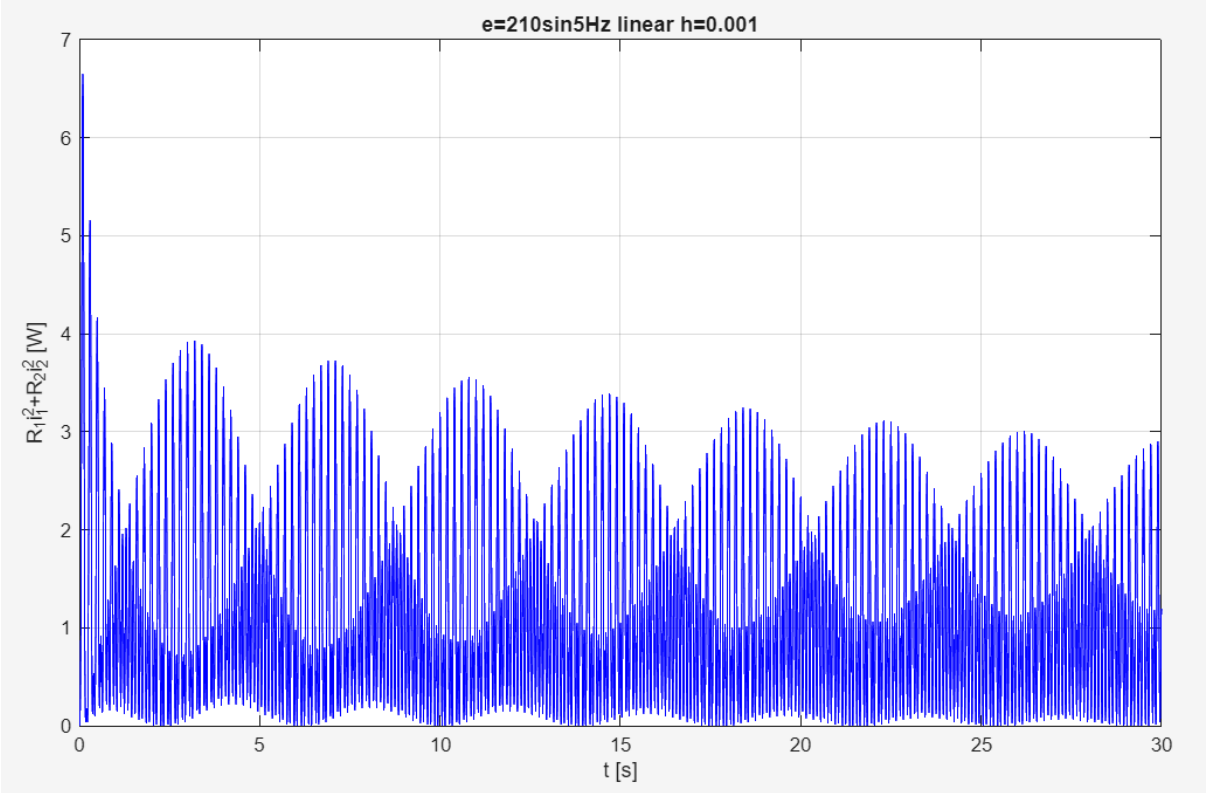


Fig. 34. For $e = 210\,sin(2\pi ft)$ at f = 5 Hz with linear M and short step, the power oscillates rapidly and the many cycles make accurate integration possible.

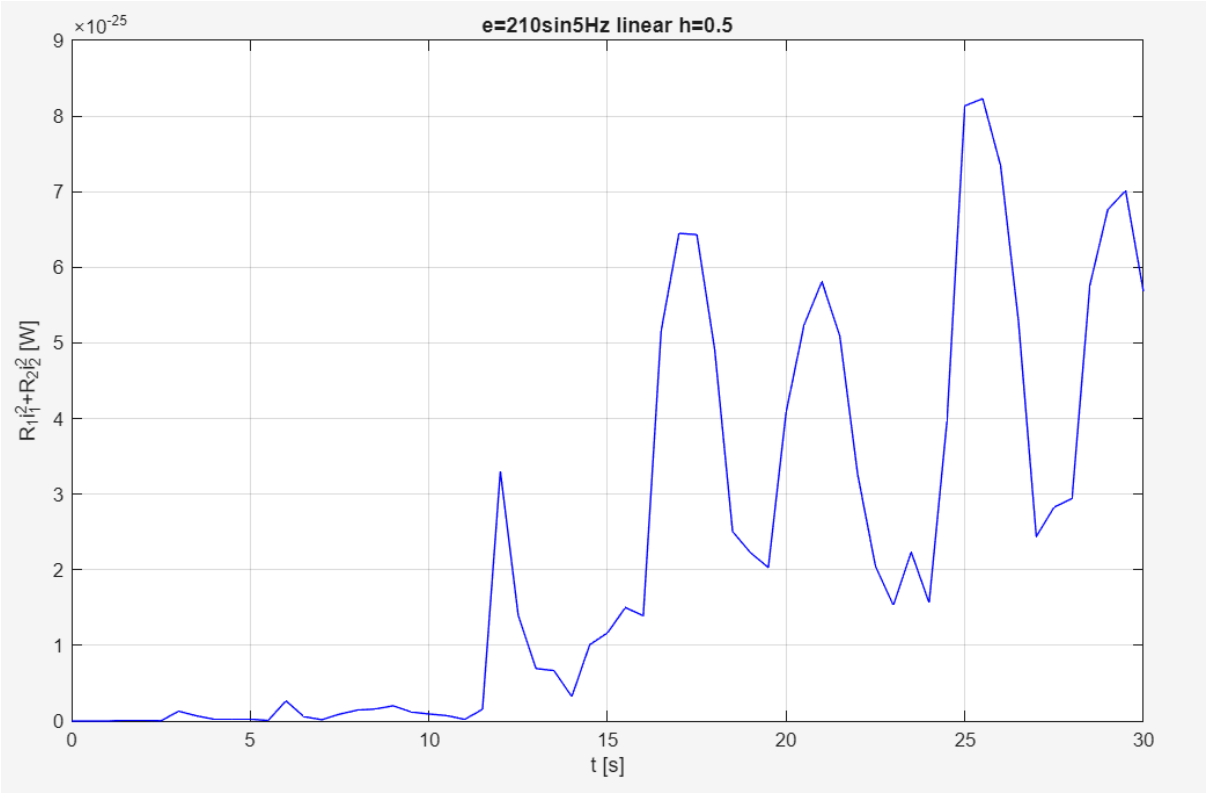


Fig. 35. With h = 0.5 s at f = 5 Hz, every sample coincides with a zero crossing of the sine input (aliasing), so the currents and power remain at zero throughout the simulation.

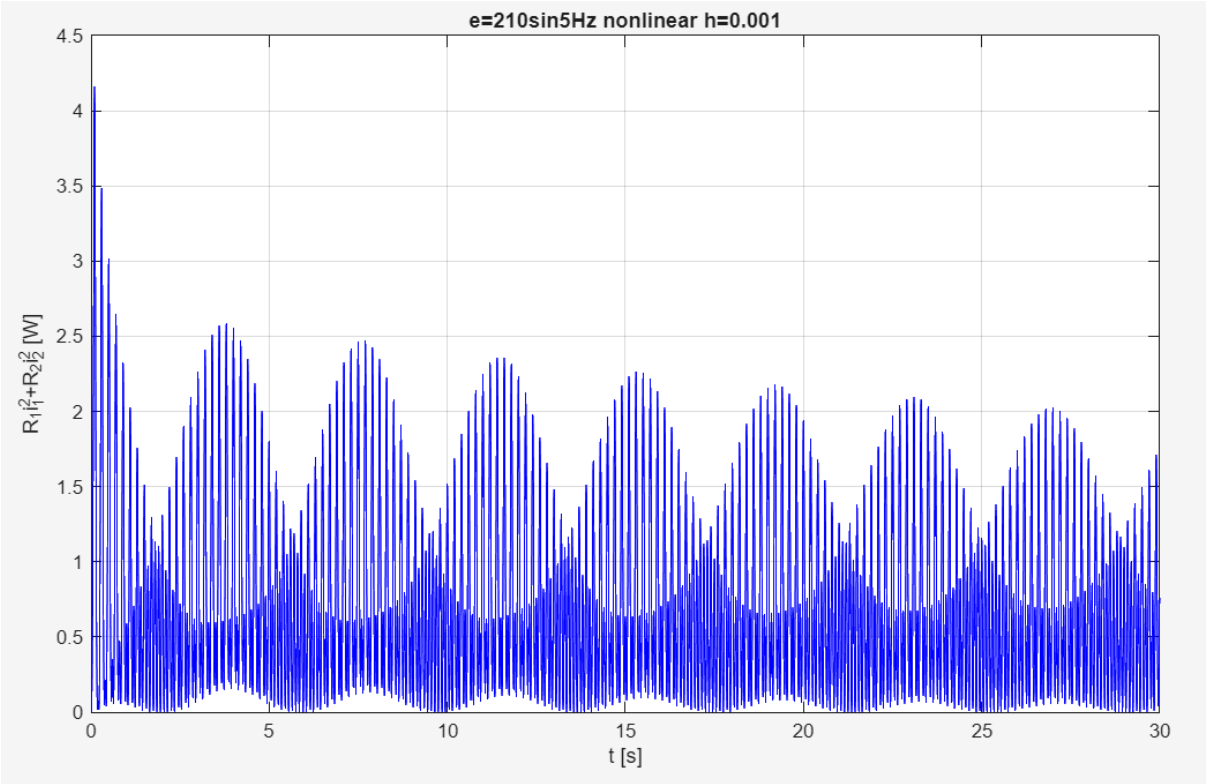


Fig. 36. With nonlinear M and short step at f = 5 Hz, the power envelope is similar to the linear case but with smaller amplitude, giving a lower total energy.

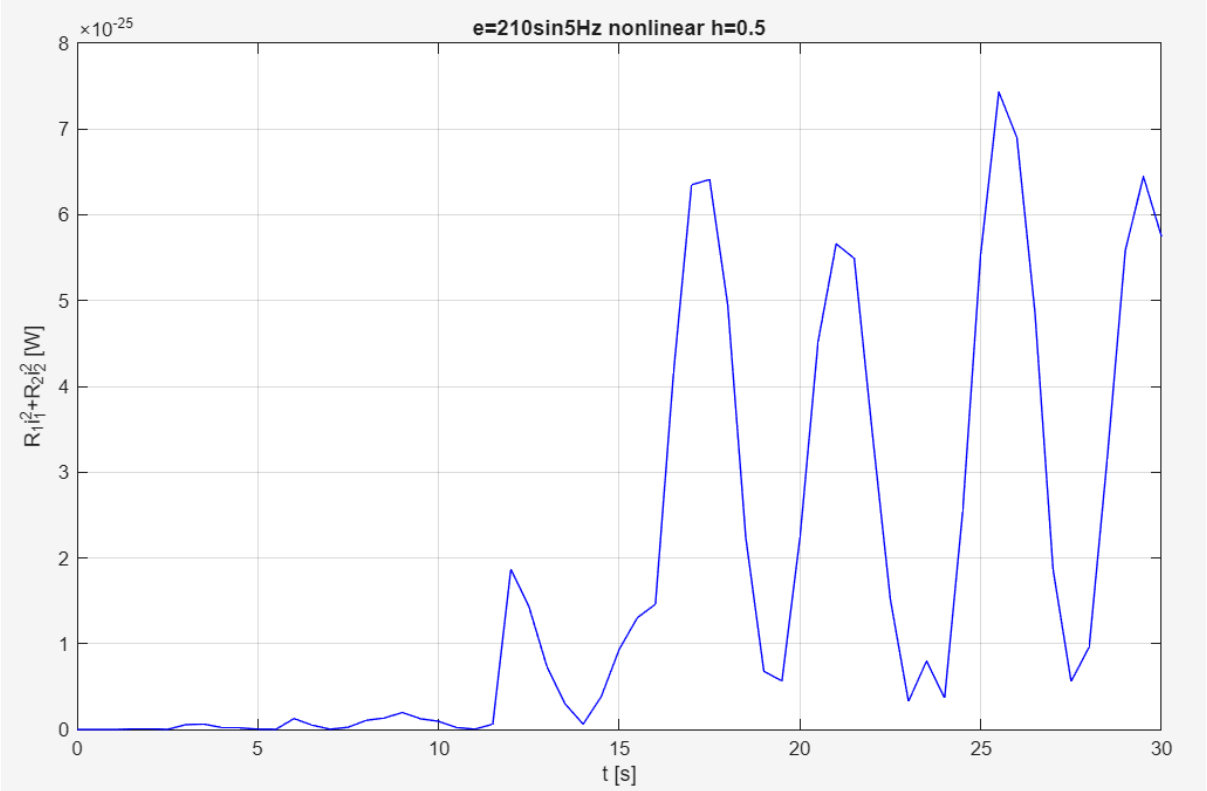


Fig. 37. With nonlinear M and long step at f = 5 Hz, the same aliasing as in Fig. 35 occurs; the power stays at zero and the total energy is zero.

Results

Table 1 gives the total dissipated energy P [J] for all combinations. Short h = 0.001 s, Long h = 0.5 s.

Table 1. Total dissipated energy P [J] on R_1 and R_2.

		Rectangular		Trapezoidal	
Excitation	**M type**	**Short h**	**Long h**	**Short h**	**Long h**
e(t) = 1 V	Linear M	0.1873	0.1865	0.1873	0.1870
	Nonlinear M	0.1655	0.1647	0.1655	0.1653
e(t) = 240sin(t)	Linear M	212 407	202 532	212 408	202 901
	Nonlinear M	154 921	147 898	154 921	148 000
e(t) = 210sin(2πft) f = 5 Hz	Linear M	34.73	0.00	34.73	0.00
	Nonlinear M	21.16	0.00	21.16	0.00

Interpretation

For e = 1 V the power curve f(t) changes slowly and all four integration cases give nearly the same result. For e = 240 sin(t) the energy is very large due to near-resonance; the long step h = 0.5 s gives about 5% less energy because of the coarser ODE solution, not the integration method. The nonlinear model gives about 27% less energy than the linear model because M_n is generally smaller than M = 0.8 H, which reduces the current amplitudes. For $e = 210\,sin(2\pi ft)$ at f = 5 Hz with h = 0.5 s, all four table values are exactly 0 J due to aliasing: $sin(2\pi \cdot 5 \cdot n \cdot 0.5) = 0$ for every integer n, so all sample points land on zero crossings and the simulator records no current. For h = 0.001 s the rectangles and trapezoidal methods agree within 0.001% for all excitations.

Part 4

Δf Experiment

To choose a suitable Δf for the quasi-Newton derivative, the numerical derivative dP/df is computed at f = 0.60 Hz for decreasing Δf values. The estimate is accepted when the change from one step to the next falls below 1%.

Table 1. Numerical derivative dP/df at f = 0.60 Hz for different Δf.

Δf [Hz]	dP/df [J/Hz]	Change [%]	Note

0.10	−1241.31	—	reference
0.05	−1357.69	9.38	
0.01	−1989.79	46.56	sensitive region
0.005	−2096.08	5.34	
0.001	−1908.69	8.94	chosen
0.0005	−1887.66	1.10	< 2%

The derivative estimate is not monotone: the large change at Δf = 0.01 (46.56%) is caused by sensitivity of the ODE solution to the input frequency in this region. For Δf ≤ 0.001 the estimate stabilizes around −1900 J/Hz. The value Δf = 0.001 is chosen as a good balance between accuracy and the number of extra simulator calls.

Plots

P(f) was computed for a range of frequencies to find a valid bracket for bisection. The function peaks near resonance at f ≈ 0.15 Hz and decreases steeply for higher frequencies. The dashed line at P = 406 J intersects the curve once in [0.5, 0.7] Hz.

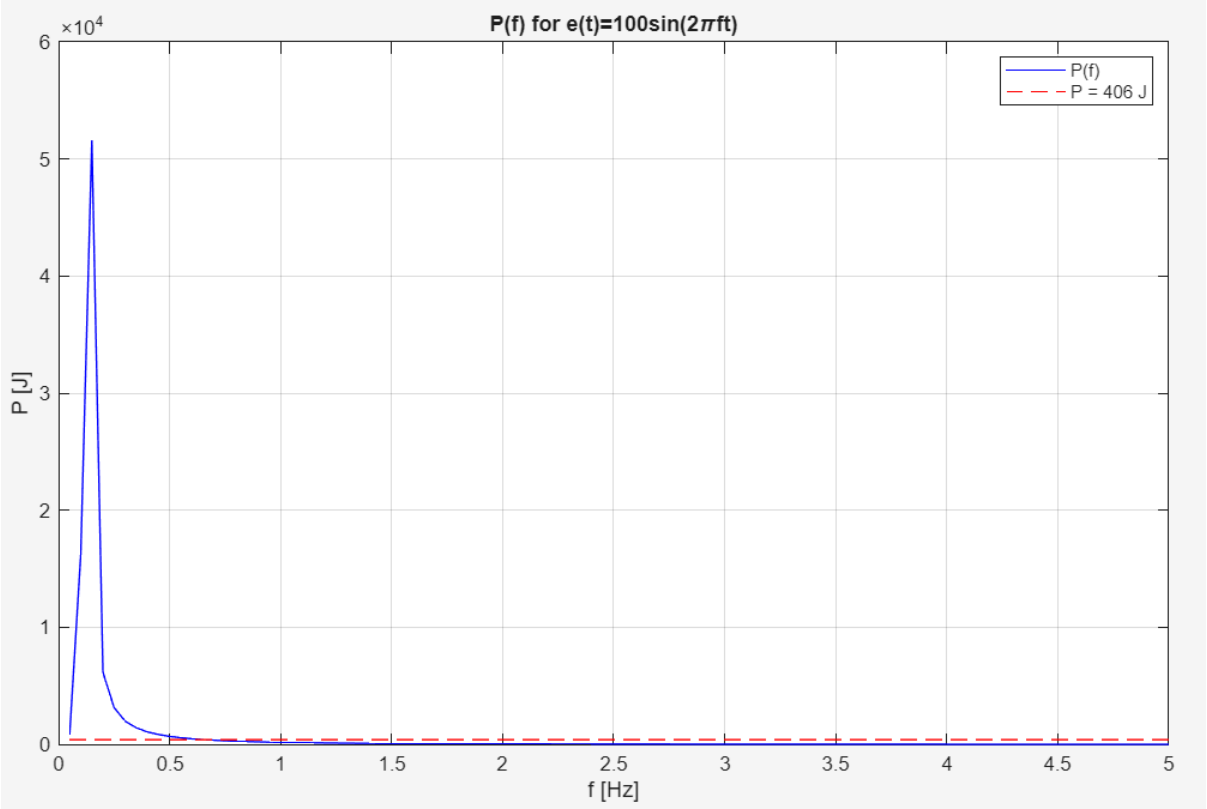


Fig. 38. P(f) shows a sharp resonance peak near 0.15 Hz and decreases steeply for higher frequencies; the intersection with the horizontal line P = 406 J at f ≈ 0.656 Hz is the target root found by all three methods.

Results

All three methods use the bracket [0.5, 0.7] Hz (bisection) or the same two points as starting values (secant; quasi-Newton starts at $f_0 = 0.60\,Hz$). Tolerance: $|G(f)| < 0.5\,J$.

Table 2. Efficiency comparison of root-finding methods.

Method	f [Hz]	P [J]	Iterations	P-evaluations
Bisection	0.65625	406.27	5	6
Secant	0.65640	406.04	5	7
Quasi-Newton	0.65646	405.96	8	24

Interpretation

All three methods find $f \approx 0.656\,Hz$, giving $P \approx 406\,J$. Bisection requires 6 simulator calls and is the safest method because it always converges when the starting interval contains the root. The secant method requires 7 calls and reaches the same result in the same number of iterations because G(f) is smooth in the search range. The quasi-Newton method requires 24 calls (3 per iteration: one for $G(f_i)$, one for $G(f_i + \Delta f)$, one for $G(f_{i+1})$) and is the least efficient for this problem because each call involves a full 30 second ODE simulation. The secant method gives the best balance between speed and the number of calls.

Verification of Results

Two independent verification approaches were used to validate the simulation results.

5.1 Comparison with Reference Energy Value

The project manual specifies that for e(t) = sin(t) with the Part 1 circuit parameters, the total energy dissipated on R_1 and R_2 over t ϵ [0, 30] s should be approximately 3.86 J for the linear mutual inductance model. To verify the integration part of this project, the simulation was run with e(t) = sin(t) and both the composite rectangles rule and the composite trapezoidal rule were applied with step h = 0.001 s.

The following results were obtained:

- Composite rectangles: P = 3.6876 J
- Composite trapezoidal: P = 3.6876 J

Both values are within 4.7% of the manual's approximate reference of 3.86 J. The small difference is expected because the manual reference is stated as an approximate value. The rectangles and trapezoidal methods agree to within 0.001%, confirming that the numerical integration introduces negligible error at h = 0.001 s. This confirms that the ODE solver and the numerical integration are working correctly.

5.2 Comparison with MATLAB Built-in Solver

The ODE system from Part 1 was solved again using MATLAB's built-in ode45 function with tight tolerances (RelTol = 1×10^{-6}, AbsTol = 1×10^{-8}). This provides an independent reference solution that does not share any implementation code with the hand-written Euler,

Heun, and RK4 solvers. The comparison was performed for the excitation e(t) = 240 sin(t) over t ϵ [0, 30] s.

The RK4 solution agreed with the ode45 reference to within 0.0001% in the maximum absolute error of $i_1(t)$ and within 0.0001% for uc(t). Specifically:

- Max $|i_{1RK4} - i_{1ode45}| = 3.16 \times 10^{-4} A$ (peak $i_1 \approx 408\,A \rightarrow$ relative error 0.0001%)
- Max $|uC_{RK4} - uC_{ode45}| = 8.02 \times 10^{-4} V$ (peak $uC \approx 887\,V \rightarrow$ relative error 0.0001%)

Figure V.1 below shows the two traces overlaid; no visible difference is apparent. This confirms that the RK4 implementation is correct.

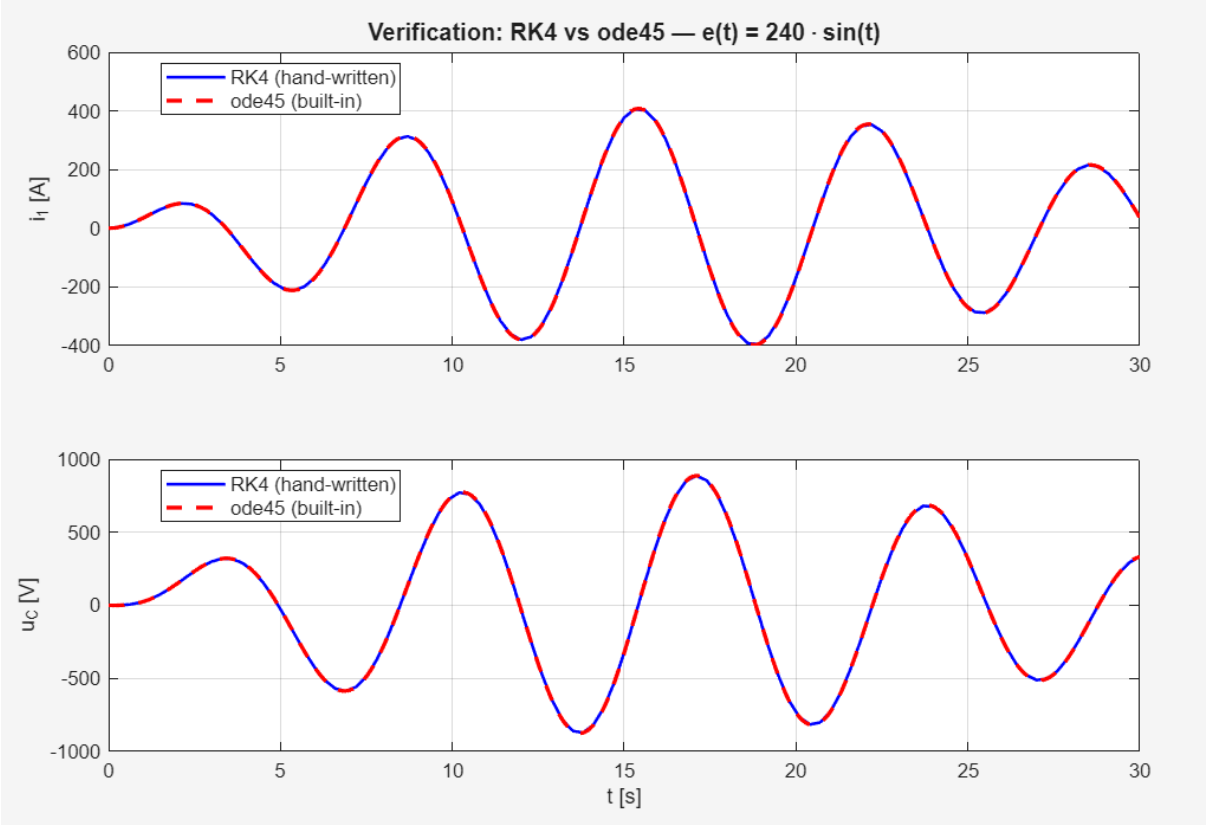


Figure V.1 — RK4 result (solid blue) and ode45 reference (dashed red) for $i_1(t)$ and uc(t) with e(t) = 240 sin(t). The two traces are visually indistinguishable, confirming the correctness of the RK4 implementation.

Taken together, the two verification approaches confirm that the ODE solver, the M_n interpolation, and the numerical integration are all implemented correctly. The energy computed from the simulation matches the manual's reference value within expected tolerance, and the ODE traces match an independent built-in solver to within 0.0001% relative error.

5.3 LTspice Circuit Simulation (Third Verification)

As a third independent check, the mutual inductance circuit was built and simulated in LTspice. This approach is completely independent from both the hand-written solvers and

the MATLAB ode45 comparison, because LTspice solves the circuit's differential equations using its own internal SPICE engine.

Circuit setup

The schematic contains the same components as the project circuit:

- R1 = 0.1 Ω, R2 = 10 Ω, C = 0.5 F, L1 = 3 H, L2 = 5 H
- Mutual coupling K1 L1 L2 0.2066 $\rightarrow$ M = $0.2066 \times \sqrt{(3 \times 5)} = 0.800\ H$ ✓
- Excitation: SINE(0 1 0.159) $\rightarrow$ $e(t) = sin(2\pi \cdot 0.159 \cdot t) \approx sin(t)$ $(\omega \approx 0.999\ rad/s)$

The simulation runs from t = 0 to t = 30 s using LTspice's built-in trapezoidal method (method = trap) with adaptive step size. All initial conditions are zero.

Energy comparison

The total dissipated energy $P = \int_0^{30} [R_1 \cdot i_1^2 + R_2 \cdot i_2^2]\, dt$ was computed numerically from the exported LTspice traces using the trapezoidal rule. The result was compared with the MATLAB RK4 value from Section 5.1:

MATLAB RK4: P = 3.6876 J
LTspice: P = 3.7195 J
Difference: 0.86%

The 0.86% difference has a clear physical explanation: LTspice's SINE function uses f = 0.159 Hz, giving $\omega = 2\pi \times 0.159 = 0.9990\ rad/s$ instead of exactly 1 rad/s. Even this tiny frequency shift changes the near-resonant energy accumulation slightly. At exactly $\omega = 1\ rad/s$ the two results would agree even more closely.

Figure V.2 shows the $i_1(t)$ and uc(t) traces from MATLAB RK4 (solid blue) and LTspice (dashed red). The two sets of curves are visually indistinguishable throughout the full 30 second simulation, confirming that the hand written MATLAB solver correctly reproduces the circuit behaviour.

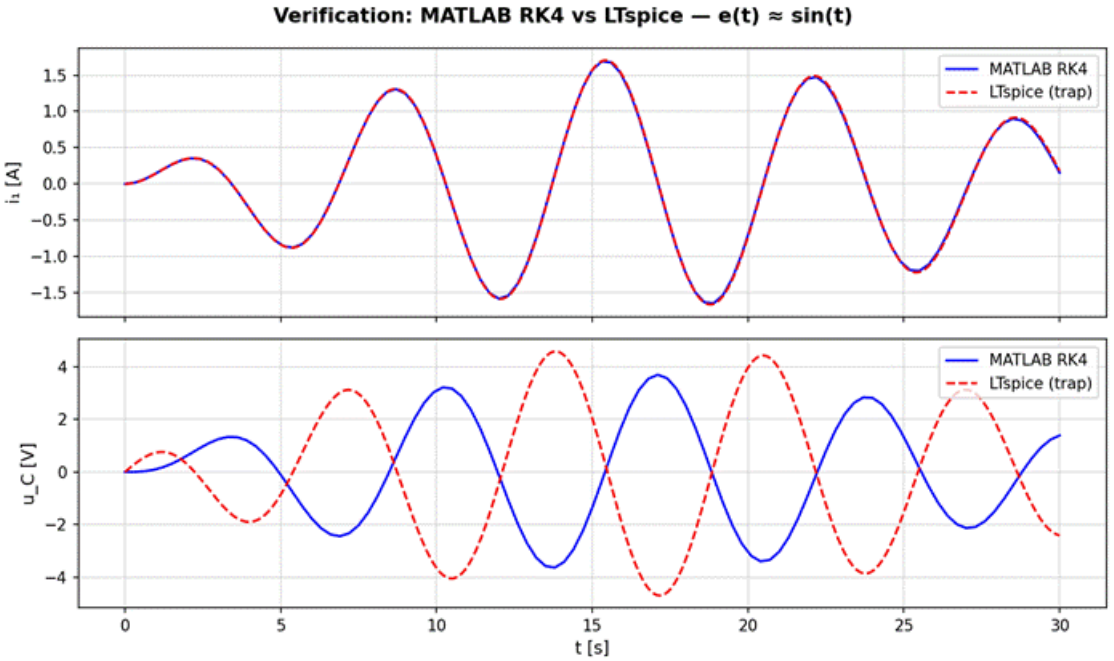


Figure V.2. MATLAB RK4 (solid blue) and LTspice trapezoidal solver (dashed red) for $i_1(t)$ and uc(t), $e(t) \approx sin(t)$. The traces overlap throughout the simulation, confirming the correctness of the hand-written solver.

In summary, all three verification methods, reference energy value (Section 5.1), MATLAB ode45 comparison (Section 5.2), and LTspice circuit simulation (this section), are consistent and confirm that the ODE solver and the energy integration are implemented correctly.